Induced Subgraphs in Ramsey Graphs

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Abstract

In this dissertation, we investigate the "random-like" features of Ramsey graphs, with an emphasis on the existence of induced subgraphs of many different sizes.

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1 Introduction

Classical Ramsey Theory asks for the minimum number of vertices required to guarantee the existence of a clique or an independent set in a graph. We refer to these maxima as *Ramsey numbers*.

Definition 1.1. The *Ramsey number* $R(s,t)$ is the minimum number N such that any graph on N vertices contains a clique of size s or an independent set of size t.

The existence of these numbers was proven by Ramsey [13] in 1930, but their quantitative behaviors remain largely unknown despite decades of intensive research. In particular, it is known that $R(3,3) = 6$ and $R(4,4) = 18$, but the exact values of $R(5,5)$ and beyond are yet to be determined. The cornerstone of Ramsey Theory due to Erdős and Szekeres in 1935 gives the following upper bound on $R(s,t)$:

Theorem 1.1 (Erdős-Szekeres [7])**.** *For $s,t \geq 2$,*

$$R(s,t) \leq \binom{s+t-2}{s-1}.$$

Following the fact that $\binom{2t-2}{t-1} < 2^{2t}$, we can see $R(t,t) < 4^t$. Very little progress had been made on this bound until the recent breakthrough of Campos, Griffiths, Morris, and Sahasrabudhe [4], giving the first exponential improvement since the result of Erdős and Szekeres. On the other hand, the lower bound of Ramsey numbers has been even harder to determine. A rather basic result which already gives the best known exponents is the famous random graph construction due to Erdős in 1947, marking one of the first applications of the probabilistic method in combinatorics. Before stating the result, we define the *Erdős-Rényi random graph*.

Definition 1.2. For $n \geq 1$ and $p \in [0,1]$, the *random graph* $G_{n,p}$ is the graph on n vertices in which each of the $\binom{n}{2}$ edges is present independently with probability p.

Remark: For constant p, the random graph $G_{n,p}$ is dense in the sense that it has $\Theta(n^2)$ many edges almost surely.

Theorem 1.2 (Erdős [8])**.** *For $t \geq 3$,*

$$\sqrt{2}^t < R(t,t) < 4^t.$$

Proof. The upper bound follows from Theorem 1.1. For lower bound, set $N = \lfloor \sqrt{2}^t \rfloor$ and consider the random graph $G_{N,1/2}$. The expected number of cliques of size t is

$$\binom{N}{t} \cdot 2^{-\binom{t}{2}} < \frac{N^t}{t!} 2^{-\binom{t}{2}} \leq \frac{\sqrt{2}^{t^2}}{t!} 2^{-\binom{t}{2}} = \frac{\sqrt{2}^t}{t!} = \frac{\sqrt{2}^{t-3}}{t(t-1)\cdots 4} \cdot \frac{\sqrt{2}^3}{3!} \leq \frac{\sqrt{2}^3}{3!} < \frac{1}{2}.$$

Similarly, the expected number of independent sets of size t is less than $1/2$. It now follows that there exists a graph on N vertices containing neither a clique nor an independent set of size t. $\square$

The above strategy may be optimized to give a lower bound of $\frac{t}{e\sqrt{2}}\sqrt{2}^t$, which has been astonishingly resilient to improvement. The only noteworthy improvement since 1947 is achieved in 1977 by Spencer [15] using the Lovász local lemma to raise the bound to $(1-o(1))\frac{t\sqrt{2}}{e}\sqrt{2}^t$, an improvement by a factor of 2. For further background on Ramsey theory, see Conlon, Fox, and Sudakov [5].

We define a shorthand for cliques and independent sets.

Definition 1.3. A set $S \subseteq V(G)$ is *homogeneous* in a graph G if it induces a clique or an independent set in G. We denote the size of the largest homogeneous set in G by $\hom(G)$.

Theorem 1.2 may therefore be restated as

$$\frac{1}{2}\log_2 n < \hom(G) < 2\log_2 n,$$

for any graph G on n vertices. Apart from determining the bounds of Ramsey numbers, the search for explicit graph constructions with $\hom(G) = O(\log n)$ has also been a major open problem in Ramsey Theory. We refer to such graphs as *Ramsey graphs*.

Definition 1.4. Let $C > 0$ be a constant. A *C-Ramsey graph* is a graph with no homogeneous set of size greater than $C \log n$.

Despite decades of intensive research, all known constructions of Ramsey graphs still rely on some randomness. Thus it is believed that Ramsey graphs inherently resemble dense random graphs like $G_{n,1/2}$, and there are a number of theorems showing that certain "richness" properties of random graphs also hold for Ramsey graphs. The first result that cemented this connection is due to Erdős and Szemerédi, showing that Ramsey graphs are also dense.

Theorem 1.3 (Erdős-Szemerédi [8])**.** *Let G be a graph on n vertices with edge density $d \le 1/2$. Then G contains a homogeneous set of size greater than $K = \frac{c}{d\log(1/d)}\log n$, for some constant $c > 0$ and sufficiently large n.*

Proof. Suppose not. Pick $c < 1/32$. Since G contains $d\binom{n}{2}$ edges, there exists $m \ge n/2$ vertices in G with degree at most dn, by the pigeonhole principle. We work with the graph G' induced by these m vertices. Let I be the largest independent set in G' and let $k = |I|$. Since $\deg(v) \le dn$ for all $v \in I$, we have $\sum_{v\in I}\deg(v) \le dnk \le 2dmk$. Put $s = 8dk$. Again by the pigeonhole principle, there are at least $m/2 - k$ vertices $v \in V(G')\setminus I$ with $\deg_I(v) < s$. Let J be the set of these vertices. Consider the family $\mathcal{F}$ of subsets of I with size less than s. Partition J by each vertex's neighbourhood in I, and note that each resulting class corresponds to an element of $\mathcal{F}$. By the standard bounds $\binom{x}{r} \le (ex/r)^r$ and $1 + x \le e^x$,

$$|\mathcal{F}| \le s\binom{k}{s-1} \le s\left(\frac{ek}{s-1}\right)^{s-1} = s\left(\frac{s}{s-1}\right)^{s-1}\left(\frac{ek}{s}\right)^{s-1} \le se\left(\frac{ek}{s}\right)^{s-1} = \frac{s^2}{k}\left(\frac{ek}{s}\right)^{s}.$$

By assumption $|I| = k < K$, and so

$$s < 8dK = \frac{8c\log n}{\log(1/d)} = O(\log n).$$

Since $s/k = 8d$, it follows that

$$|\mathcal{F}| \leq \frac{s^2}{k}\left(\frac{ek}{s}\right)^s \leq O(\log n)\cdot\left(\frac{e}{8}\cdot\frac{1}{d}\right)^s \leq O(\log n)\cdot\left(\frac{1}{d}\right)^{\frac{8c\log n}{\log(1/d)}} = O(\log n)\cdot n^{8c} < n^{1/3}.$$

But then

$$\frac{|J|}{|\mathcal{F}|} \geq \frac{m/2-k}{n^{1/3}} \geq \sqrt{n},$$

so by the pigeonhole principle, there exists a class of vertices Z in J of size at least $\sqrt{n}$. Let $G'' = G[Z]$. Note that G'' does not contain an independent set of size s, otherwise exchange it with Y in I creates a larger independent set in G', contradicting the maximality of I. Let t be the size of the largest clique in G'', and note that $t < K$ by assumption. The Erdős-Szekeres and the standard binomial bounds give

$$\sqrt{n} \leq |Z| \leq \binom{s+K-2}{s-1} \leq \binom{s+K}{s} \leq \binom{8dK+K}{8dK} \leq \left(\frac{e(8d+1)}{8d}\right)^s \leq \left(\frac{2}{d}\right)^s,$$

where the last inequality holds because $e(1+8d)/8 \leq 5e/8 \leq 2$ for $d \leq 1/2$. But then since $d \leq 1/2$ and $c < 1/32$,

$$\left(\frac{2}{d}\right)^s \leq \left(\frac{2}{d}\right)^{\frac{8c\log n}{\log(1/d)}} = n^{8c\cdot\frac{\log(2/d)}{\log(1/d)}} = n^{8c\left(1+\frac{\log 2}{\log(1/d)}\right)} \leq n^{16c} < \sqrt{n},$$

contradiction. □

Recall that the Erdős-Szekeres theorem (Theorem 1.1) gives $\hom(G) \geq \frac{1}{2}\log_2 n$, regardless of the edge density. Theorem 1.3 improves this when the density is small. For example, if $d = 1/n^\varepsilon$ for some $\varepsilon > 0$, then $\hom(G) = \Omega\left(\frac{n^\varepsilon}{\log n^\varepsilon}\log n\right)$, vastly larger than $\frac{1}{2}\log_2 n$. Intuitively, a sparse graph should contain a large independent set, and the theorem makes this notion precise.

If G is C-Ramsey, then applying Theorem 1.3 to whichever G or $\overline{G}$ has density $d \leq 1/2$ yields $c/(d\log(1/d)) \leq C$, forcing d to be bounded away from 0. By the same argument applied to $\overline{G}$ forces d to be bounded away from 1. This gives the following corollary.

Corollary 1.4. *For any C-Ramsey graph G, there exists $\delta = \delta(C) \in (0, 1/2)$ such that the edge density of G lies in the interval $[\delta, 1-\delta]$.*

Due to the rich variety of induced subgraphs in dense random graphs, a popular angle to study the pseudorandom properties of Ramsey graphs is to consider their induced subgraphs. Some notable results are a theorem of Shelah [14] which asserts

that every n-vertex Ramsey graph contains $2^{\Omega(n)}$ non-isomorphic induced subgraphs, and a theorem of Prömel and Rödl [12] which shows the existence of $\delta = \delta(C) > 0$ such that every C-Ramsey graph on n vertices contains an induced copy of all graphs on at most $\delta \log n$ vertices.

It is also interesting to look at the edge statistics of induced subgraphs in Ramsey graphs. For a graph G, let

$$\Phi(G) \coloneqq \{e(H) : H \text{ is an induced subgraph of } G\}$$

Erdős and McKay [6] conjectured that for any Ramsey graph G on n vertices, there exists $\delta = \delta(C) > 0$ such that

$$\{1, 2, \ldots, \delta n^2\} \subseteq \Phi(G).$$

It was proven for Random graphs by Calkin, Frieze, and Mckay [3], but the conjecture for Ramsey graphs remained open for decades and only started to show significant progress in recent years. In particular, a chain of results were sparked after the work of Narayanan, Sahasrabudhe, and Tomon [11] establishing that $|\Phi(G)| = n^{2-o(1)}$ for Ramsey graph G on n vertices. Their proof builds on some ideas from the work of Bukh and Sudakov [2] which showed that every Ramsey graph contains a linear sized subgraph with many distinct degrees, a feature that dense random graphs also exhibit. Kwan and Sudakov [9] later improved the bound to $|\Phi(G)| = \Omega(n^2)$ by refining the argument of Narayanan, Sahasrabudhe, and Tomon. Eventually, Kwan, Sah, Sauermann, and Sawhney [10] closed the conjecture via Fourier-analytic techniques on graph polynomials.

This dissertation will start by studying the result of Bukh and Sudakov [2] on large induced subgraphs in Ramsey graphs with many distinct degrees. This work not only tightens the connection between Ramsey graphs and random graphs, but also builds the foundation for the subsequent $|\Phi(G)| = \Omega(n^2)$ result by Kwan and Sudakov which we will present in the following chapter. Lastly, we go through the broad strokes of the proof to the Erdős-McKay conjecture by Kwan, Sah, Sauermann, and Sawhney [10], touching on the core ideas of their approach.

1.1 The anticoncentration principle

The core results of this dissertation surrounds a key probabilistic mechanism called *anticoncentration*. In the following chapters, we will repeatedly sample a random vertex set $U \subseteq V(G)$ and study the induced subgraph $G[U]$. A natural question to ask in this setting is how likely it is for two vertices $x, y \in V(G)$ to have the same degree in $G[U]$?

Observe that $\deg_U(x) - \deg_U(y)$ counts only vertices in the symmetric difference $D = N_G(x) \Delta N_G(y)$. Write $A = N(x) \setminus N(y)$ and $B = N(y) \setminus N(x)$ and we have

$$\deg_U(x) - \deg_U(y) = |U \cap A| - |U \cap B|, \tag{1}$$

and so x, y have the same degree in $G[U]$ precisely when $|U \cap A| = |U \cap B|$.

Suppose first that U is formed by including each vertex independently with probability p. Then each vertex $v \in A$ independently contributes $+1$ to the difference $|U \cap A| - |U \cap B|$ with probability p, while each vertex $v \in B$ independently contributes -1 to the difference with probability p. The difference is thus a sum of $|D| = |A| + |B|$ independent bounded random variables. When $|D|$ is large, say $|D| \geq \varepsilon n$, this sum has standard deviation $\Theta(\sqrt{n})$. Since $|U \cap A| - |U \cap B|$ is integer-valued, we can show the probability that it equals any particular integer is at most $O(1/\sqrt{n})$ via the one-dimensional Erdős-Littlewood-Offord theorem. Thus we conclude that $\mathbb{P}[\deg_U(x) = \deg_U(y)] = O(1/\sqrt{n})$. We will formalize this with greater detail in Chapter 3.

If instead U is a uniformly random subset of a fixed size m, the vertices of U are no longer independent. Nevertheless, conditioning on $k = |U \cap D|$, we may observe that, given k, the split of these k elements between A and B follows a hypergeometric distribution. With the same assumption that $|D|$ is large, Stirling's approximation applied to the hypergeometric probability then yields the same $O(1/\sqrt{n})$ bound. We will carry out this calculation in detail in Chapter 2.

An intuition to the bound $O(1/\sqrt{n})$ is that, in either sampling method, the difference $\deg_U(x) - \deg_U(y)$ is approximately normally distributed with standard deviation $\Theta(\sqrt{n})$. Since the probability density function $\phi(x)$ of a normal distribution with standard deviation σ is at most $1/(\sigma\sqrt{2\pi})$ for all x, the probability that $\deg_U(x) - \deg_U(y)$ equals any particular value is $O(1/\sigma) = O(1/\sqrt{n})$, and this principle is the key to the results in Chapter 2 and Chapter 3. The resolution of the Erdős-McKay conjecture in Chapter 4 requires a substantially more powerful approach via Fourier analysis, but the underlying idea remains: when a random variable is approximately normal with large standard deviation, its point probabilities are small.

1.2 Notations and tools

We call $e(G)$ the *size* of a graph G. We omit all floor and ceiling signs when they are not crucial for simplicity.

Proposition 1.5 (Turán's theorem)**.** *Every graph G with n vertices and m edges contains an independent set of size at least $n^2/(2m+n)$.*

Proposition 1.6 (Hoeffding's inequality)**.** *Let $X_1, X_2, \ldots, X_n$ be independent real-valued random variables with $0 \leq X_i \leq 1$ for each i, and write $X = \sum_{i=1}^n X_i$. Then for $t \geq 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq 2e^{-2t^2/n}.$$

Proposition 1.7. *Let X denote the number of successes in D draws, without replacement, from a population of size N which contains M successes. Then $\mathbb{E}[X] = DM/N$ and for $t \geq 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq t) \leq 2e^{-2t^2 D}.$$

Proof. $\square$

Definition 1.5 (Littlewood-Offord type random variable)**.** A random variable X is of (n, p)-*Littlewood-Offord type* if it is of the form $X = a_1\xi_1 + a_2\xi_2 + \cdots + a_n\xi_n + C$, where $a_1, a_2, \ldots, a_n \in \mathbb{Z} \setminus \{0\}$ and $C \in \mathbb{Z}$ are fixed and $\xi_1, \xi_2, \ldots, \xi_n$ are independent Bernoulli random variables with parameter p.

Proposition 1.8 (Littlewood-Offord theorem)**.** *Let $p > 0$ be constant and X be a random variable of (n, p)-Littlewood-Offord type. Then for any $t \in \mathbb{Z}$,*

$$\mathbb{P}[X = t] = O(1/\sqrt{n}).$$

2 Subgraphs with Many Degrees

In this chapter, we present the following result of Bukh and Sudakov.

Theorem 2.1 (Bukh-Sudakov [2])**.** *Every Ramsey graph on n vertices induces a subgraph on $\Omega(n)$ vertices with $\Omega(\sqrt{n})$ distinct degrees.*

This was originally conjectured by Erdős, Faudree, and Sós [6] in light of the observation that the random graph $G_{n,1/2}$ exhibits such a property almost surely. In particular, this theorem already has implications for the edge statistics of Ramsey graphs G, as the subgraphs $G - \{v\}$ for $v \in V(G)$ have at least $\Omega(\sqrt{n})$ different sizes. This idea lays the foundation for the proof in the next chapter.

To prove Theorem 2.1, Bukh and Sudakov introduced the notion of *diversity*, which captures the idea that the neighbourhoods of most pairs of vertices are not too similar.

Definition 2.1. An n-vertex graph G is (ε, δ)-*diverse* if for $x \in V(G)$, we have $|N(x)\Delta N(y)| \geq \varepsilon n$ for all but at most n^δ vertices $y \in V(G)$.

Notice that (ε, δ)-diversity directly conditions our assumption that $|N(x)\Delta N(y)| = D \geq \varepsilon n$ in the discussion of the anticoncentration principle in Section 1.1. In particular, we may show in a diverse graph that $\mathbb{P}[\deg_U(x) = \deg_U(y)] = O(1/\sqrt{n})$ for all but at most n^δ pairs of vertices $x, y \in V(G)$, from which we use Jensen's inequality to conclude that there are many distinct degrees in a diverse graph. The following lemma formalizes this observation and reduces Theorem 2.1 to showing every Ramsey graph contains a diverse subgraph of linear size.

Lemma 2.2. *Let $\delta \in (0, 1/2)$ and let G be an (ε, δ)-diverse graph on n vertices for large enough n. Then for $n/4 \leq m \leq 3n/4$, there is a subgraph on m vertices containing vertices of $\sqrt{\varepsilon n}/10$ distinct degrees.*

Proof. Fix m as above. Let U be chosen uniformly at random among all subsets of $V(G)$ of size m. It suffices to show that with positive probability, $G[U]$ contains $\Omega(\sqrt{n})$ vertices of distinct degrees.

For distinct $x, y \in V(G)$, define $\mathbb{1}_{xy}$ to be the indicator for the event that $x, y \in V(H)$ and $\deg_U(x) = \deg_U(y)$. Then $\mathbb{E}[\mathbb{1}_{xy}] \leq \mathbb{P}[\deg_U(x) = \deg_U(y)]$. Put $D = N_G(x)\Delta N_G(y)$, and $k = |D \cap U|$. Write $A = N_G(x) \setminus N_G(y)$ and $B = N_G(y) \setminus N_G(x)$. Then

$$\mathbb{P}[\deg_U(x) = \deg_U(y)] = \mathbb{P}\left[|A \cap U| = |B \cap U|\right].$$

Suppose $|D| \geq \varepsilon n$. Since $\frac{|D|}{4} \leq \frac{m|D|}{n} \leq \frac{3|D|}{4}$, Proposition 1.7 yields

$$\mathbb{P}\left[k < \frac{|D|}{8} \text{ or } k > \frac{7|D|}{8}\right] \leq \mathbb{P}\left[|k - \mathbb{E}[k]| > \frac{|D|}{8}\right] \leq e^{-\Theta(|D|)} = e^{-\Theta(n)} = o(n^{-1/2}).$$

On the other hand, recall that U is a uniformly random subset of a fixed size m. Since every k-subset of D is equally likely to be $U \cap D$ by symmetry, conditioning on

$|U \cap D| = k$, the k elements of U that land in D form a uniformly random subset of D of size k. It follows that when conditioning on k, the event $|A \cap U| = |B \cap U| = k/2$ follows the hypergeometric distribution $\mathrm{Hyp}(|D|, |A|, k)$. Thus,

$$\mathbb{P}\left[\deg_U(x) = \deg_U(y) \,\middle|\, \frac{|D|}{8} \le k \le \frac{7|D|}{8}\right] = \frac{\binom{|A|}{k/2}\binom{|B|}{k/2}}{\binom{|D|}{k}}.$$

Since $|A| + |B| = |D|$, by the log-concavity of the binomial coefficients we have $\binom{|A|}{k/2}\binom{|B|}{k/2} \le \binom{|D|/2}{k/2}^2$. It now follows from Stirling's formula and $k(|D| - k) \ge \frac{7}{64}|D|^2$ that the above probability does not exceed

$$\frac{\binom{|D|/2}{k/2}^2}{\binom{|D|}{k}} = (1 + o(1))\sqrt{\frac{2|D|}{\pi k(|D| - k)}} \le \frac{3}{\sqrt{|D|}} \le \frac{3}{\sqrt{\varepsilon n}}. \tag{2}$$

Thus if $|N_G(x) \Delta N_G(y)| \ge \varepsilon n$ then $\mathbb{E}[\mathbb{1}_{xy}] \le 3/\sqrt{\varepsilon n} + o(n^{-1/2}) \le 4/\sqrt{\varepsilon n}$.

Now consider the degree collision graph H on U, where $x, y \in U$ are adjacent if and only if $\deg_U(x) = \deg_U(y)$. Since G is (ε, δ)-diverse, $\mathbb{E}[\mathbb{1}_{xy}] \le 4/\sqrt{\varepsilon n}$ for all but at most $n^{1+\delta} < n^{3/2}$ pairs $x, y \in V(G)$. Thus the expected number of edges in H is

$$\mathbb{E}[e(H)] = \sum_{x,y \in V(G)} \mathbb{E}[\mathbb{1}_{xy}] \le n^{1+\delta} + \frac{4}{\sqrt{\varepsilon n}} \cdot n^2 \le \frac{5}{\sqrt{\varepsilon}} n^{3/2}.$$

It now follows from Turan's Theorem (Proposition 1.5) that H contains an independent set $I \subseteq U$ of size at least

$$|I| \ge \frac{n^2}{2\mathbb{E}[e(H)] + n} \le \frac{\sqrt{\varepsilon n}}{10}.$$

In particular, each vertex in I has a distinct degree in $G[U]$. □

Remark. The hypergeometric distribution $\mathrm{Hyp}(|D|, |A|, k)$ has variance

$$\sigma^2 = \frac{k|A||B|(|D| - k)}{|D|^2(D - 1)} \le \frac{k(|D| - k)}{4(|D| - 1)},$$

where inequality follows from AM-GM. When k is not the tail end of the distribution, we have $\sigma = \Theta(\sqrt{D}) = \Theta(\sqrt{n})$, and the Stirling bound in (2) is asymptotically $1/(\sigma\sqrt{2\pi})$ as $n \to \infty$, which is exactly the maximum density of a normal distribution with standard deviation σ. This confirms the intuition from Section 1.1 that the degree difference is approximately normally distributed so its point probabilities are $O(1/\sigma) = O(1/\sqrt{n})$.

It remains to show that every Ramsey graph contains a diverse subgraph of linear size. Instead of directly proving this, we prove the same statement on a strengthening of diversity called *richness*.

Definition 2.2. An n-vertex graph G is (ε, δ)-*rich* if, for $W \subseteq V(G)$ with $|W| \geq \delta n$, we have $|N(v) \cap W| \geq \varepsilon |W|$ and $|\overline{N(v)} \cap W| \geq \varepsilon |W|$ for all but at most n^δ vertices $v \in V(G)$.

The key property of richness is that it ensures the intersection of vertex neighbourhoods with arbitrary large vertex sets are not extremely large or small. In particular, when W is the neighborhood of another vertex, richness ensures both sides of the symmetric difference $|N(x) \Delta N(y)|$ are large, rather than merely the symmetric difference itself being large. This is why richness is a stronger condition than diversity.

Lemma 2.3. *Let an n-vertex graph G be (ε, δ)-rich, with $\delta \leq 1/2$. Then G is $(\varepsilon/2, \delta)$-diverse.*

Proof. Note that for each $x \in V(G)$, either $|N(x)| \geq n/2$ or $|\overline{N(x)}| \geq n/2$. If $|N(x)| \geq n/2$, then richness implies that

$$|\overline{N(y)} \cap N(x)| \geq \varepsilon |N(x)| \geq \varepsilon n/2$$

for at most n^δ vertices $y \in V(G)$. Similarly, if $|\overline{N(x)}| \geq n/2$, then

$$|N(y) \cap \overline{N(x)}| \geq \varepsilon |\overline{N(x)}| \geq \varepsilon n/2$$

for at most n^δ vertices $y \in V(G)$. But then in either case,

$$|N(x) \Delta N(y)| = |\overline{N(y)} \cap N(x)| + |N(y) \cap \overline{N(x)}| \geq \varepsilon n/2.$$

□

We now show that every Ramsey graph contains a rich subgraph of linear size.

Lemma 2.4. *For $\delta \in (0, 1)$, there exists $\varepsilon = \varepsilon(C, \delta) > 0$ such that every C-Ramsey graph G on n vertices contains an (ε, δ)-rich subgraph on $\Omega(n)$ vertices.*

Proof. Suppose not. Let $K = K(C)$ be a large constant to be determined. Let $c = 4^{-K}$ and $\varepsilon = 1/K$. We inductively construct a sequence of disjoint vertex sets $S_1, S_2, \ldots, S_K$ of size $s = (cn)^\delta/2$, such that for all S_i one of the following conditions holds:

(i) $d(S_i, S_j) < 4\varepsilon$ for $j > i$,

(ii) $d(S_i, S_j) > 1 - 4\varepsilon$ for $j > i$.

Alongside, we also construct a sequence of induced subgraphs $G = G[U_0] \supset G[U_1] \supset \cdots \supset G[U_K]$, such that $|U_i| \geq (\delta/4)|U_{i-1}|$ and $S_i \subseteq U_{i-1}$ for all i.

We first show how this construction gives the lemma. By working with $\overline{G}$ instead of G if necessary, we may assume that $r \geq K/2$ of the sets $S_1, \ldots, S_K$ satisfy condition

(i), say $S_{i_1}, \dots, S_{i_r}$. Put $S = \bigsqcup_{k=1}^r S_{i_k}$. Since each S_{i_k} has size s and $\varepsilon = 1/K$,

$$d(S) = \frac{\sum_{k=1}^r e(S_{i_k})}{\binom{|S|}{2}} + \frac{\sum_{k<l} e(S_{i_k}, S_{i_l})}{\binom{|S|}{2}} \le \frac{r\binom{s}{2}}{\binom{rs}{2}} + \max_{(k,l)} d(S_{i_k}, S_{i_l}) \le \frac{1}{r} + 4\varepsilon \le \frac{6}{K}.$$

The Erdős–Szemerédi theorem (Theorem 1.3) now yields a homogeneous set $A \subseteq S$ of size

$$|A| \ge \frac{\gamma K}{10\log(K/10)} \log|S| = \frac{\gamma\delta K}{10\log(K/10)} \log cn = \frac{\gamma\delta K}{10\log(K/10)} \log n + o(\log n),$$

for some absolute constant $\gamma > 0$ given by the theorem. Take K to be large enough so that $\frac{\gamma\delta K}{10\log(K/10)} > C$. Then $|A| > C\log n$, contradiction.

We now construct the sets S_i and U_i inductively. Suppose $U_1, \dots, U_{i-1}$ and $S_1, \dots, S_{i-1}$ have been constructed. By induction, $|U_{i-1}| \ge (\delta/4)^{i-1}|U_0| \ge cn$. Since $G[U_{i-1}]$ is not (ε, δ)-rich by assumption, U_{i-1} contains subsets W_i, Y_i with $|W_i| \ge \delta|U_{i-1}|$ and $|Y_i| = (cn)^\delta$, such that for each $y \in Y_i$ we have $|N_{G_{i-1}}(y) \cap W_i| < \varepsilon|W_i|$ or $|\overline{N_{G_{i-1}}(y)} \cap W_i| < \varepsilon|W_i|$. We consider two cases.

Suppose $|N_{G_{i-1}}(y) \cap W_i| < \varepsilon|W_i|$ for half the vertices $y \in Y_i$, and let S_i be the corresponding subset of Y_i, so $|S_i| = (cn)^\delta/2$. Put $T_i = W_i \setminus S_i$ and note that $|T_i| \ge |W_i|/2$, as $|S_i| \le |W_i|/2$ for large enough n. Then for $y \in S_i$,

$$d(y, T_i) = \frac{|N_{G_{i-1}}(y) \cap T_i|}{|T_i|} \le \frac{|N_{G_{i-1}}(y) \cap W_i|}{|T_i|} < \frac{\varepsilon|W_i|}{|W_i|/2} = 2\varepsilon.$$

Partition T_i into $U_i \sqcup Z_i$ where $U_i = \{v \in T_i : d(v, S_i) \le 4\varepsilon\}$. We show that $|U_i| \ge (\delta/4)|U_{i-1}|$ by double counting $e(T_i, S_i)/|S_i|$. Since $d(v, S_i) > 4\varepsilon$ for $v \in Z_i$,

$$\frac{e(T_i, S_i)}{|S_i|} = \sum_{v \in T_i} d(v, S_i) \ge \sum_{v \in Z_i} d(v, S_i) > 4\varepsilon|Z_i|.$$

On the other hand, since $d(y, T_i) < 2\varepsilon$ for $y \in S_i$,

$$\frac{e(T_i, S_i)}{|S_i|} = \frac{1}{|S_i|} \sum_{y \in S_i} e(y, T_i) = \frac{|T_i|}{|S_i|} \sum_{y \in S_i} d(y, T_i) < 2\varepsilon|T_i|.$$

Combining both inequalities yields $|Z_i| < |T_i|/2$. Therefore

$$|U_i| = |T_i| - |Z_i| > \frac{|T_i|}{2} \ge \frac{|W_i|}{4} \ge \frac{\delta}{4}|U_{i-1}|.$$

For $j > i$, since $S_j \subseteq U_j \subseteq U_i$, we have $d(v, S_i) \le 4\varepsilon$ for every $v \in S_j$, and averaging gives $d(S_i, S_j) \le 4\varepsilon$. Thus S_i satisfies condition (i).

Otherwise, $|\overline{N_{G_{i-1}}(y)} \cap W_i| < \varepsilon|W_i|$ for half the vertices $y \in Y_i$. Let S_i be the corresponding subset of Y_i and put $T_i = W_i \setminus S_i$, so again $|T_i| \ge |W_i|/2$ and $|S_i| = (cn)^\delta/2$. For $y \in S_i$,

$$1 - d(y, T_i) = \frac{|\overline{N_{G_{i-1}}(y)} \cap T_i|}{|T_i|} \le \frac{|\overline{N_{G_{i-1}}(y)} \cap W_i|}{|T_i|} < \frac{\varepsilon|W_i|}{|W_i|/2} = 2\varepsilon,$$

so $d(y, T_i) > 1 - 2\varepsilon$ for every $y \in S_i$. Partition T_i into $U_i \sqcup Z_i$ where $U_i = \{v \in T_i : d(v, S_i) \geq 1 - 4\varepsilon\}$. We double count $|T_i| - e(T_i, S_i)/|S_i| = \sum_{v \in T_i}(1 - d(v, S_i))$. Since $1 - d(v, S_i) > 4\varepsilon$ for $v \in Z_i$,

$$|T_i| - \frac{e(T_i, S_i)}{|S_i|} = \sum_{v \in T_i}(1 - d(v, S_i)) \geq \sum_{v \in Z_i}(1 - d(v, S_i)) > 4\varepsilon|Z_i|.$$

On the other hand, using $d(y, T_i) > 1 - 2\varepsilon$ for $y \in S_i$,

$$|T_i| - \frac{e(T_i, S_i)}{|S_i|} = |T_i| - \frac{|T_i|}{|S_i|}\sum_{y \in S_i} d(y, T_i) < |T_i| - (1 - 2\varepsilon)|T_i| = 2\varepsilon|T_i|.$$

Combining both inequalities gives $|Z_i| < |T_i|/2$, hence $|U_i| > |T_i|/2 \geq (\delta/4)|U_{i-1}|$ as in the other case. For $j > i$, each $v \in S_j \subseteq U_i$ satisfies $d(v, S_i) \geq 1 - 4\varepsilon$, so $d(S_i, S_j) \geq 1 - 4\varepsilon$. Thus S_i satisfies condition (ii). This completes the induction. $\square$

Theorem 2.1 now follows by a simple application of Lemmas 2.4, 2.3, and 2.2 in sequence.

Proof of Theorem 2.1. Let G be a C-Ramsey graph on n vertices, and fix any $\delta \in (0, 1/2)$. By Lemma **??**, there exists $\varepsilon = \varepsilon(C, \delta) > 0$ and an (ε, δ)-rich induced subgraph $G[V'] \subseteq G$ with $|V'| \geq cn$ for some $c = c(C, \delta) > 0$. Put $n' = |V'|$. By Lemma **??**, $G[V']$ is $(\varepsilon/2, \delta)$-diverse on n' vertices. Applying Lemma **??** to $G[V']$ with $m = n'/2$, we obtain an induced subgraph of $G[V']$ on $m \geq cn/2 = \Omega(n)$ vertices containing $\frac{\sqrt{\varepsilon n'/2}}{10} \geq \frac{\sqrt{\varepsilon cn/2}}{10} = \Omega(\sqrt{n})$ vertices of distinct degrees.

3 Subgraphs of Quadratically Many Sizes

Recall that Erdős and Mckay conjectured that for any Ramsey graph G on n vertices, there exists $\delta = \delta(C) > 0$ such that

$$\{1, 2, \ldots, \delta n^2\} \subseteq \Phi(G) := \{e(H) : H \text{ is an induced subgraph of } G\},$$

That is, for any $1 \leq t \leq \delta n^2$, there exists an induced subgraph with exactly t edges. The conjecture was widely open for decades. In 2003, Alon, Krivelevich and Sudakov [1] proved the conjecture with n^δ in place of δn^2, but little progress was made afterwards.

Theorem 3.1 (Alon, Krivelevich and Sudakov [1])**.** *For any $C > 0$, there exists $\delta = \delta(C) > 0$ such that*

$$\{1, 2, \ldots, n^\delta\} \subseteq \Phi(G).$$

Since the conjecture implies $|\Phi(G)| = \Omega(n^2)$ and the triavial upper bound gives $|\Phi(G)| \leq \binom{n}{2}$, a natural relaxation of the conjecture is to ask whether $\Phi(G)$ indeed has quadratic size asymptotically. In 2017, Narayanan, Sahasrabudhe, and Tomon [11] proved a near optimal bound of $|\Phi(G)| = n^{2-o(1)}$ by building on the notion of diversity from Chapter 2. The desired quadratic bound was achieved by Kwan and Sudakov [9] a few years later, and this chapter is dedicated to presenting their proof.

Theorem 3.2 (Kwan and Sudakov [9])**.** *For any $C > 0$,*

$$|\Phi(G)| = \Theta(n^2).$$

Recall that Theorem 2.1 provides a bound of $|\Phi(G)| = \Omega(\sqrt{n})$ for any Ramsey graph G by removing a single vertex to construct each subgraph. Two ideas may be used to improve this construction. First, we can apply the same idea to many different induced subgraphs rather than merely G itself. Second, instead of deleting only one vertex, we may add and remove various combinations of vertices to produce many more distinct sizes.

Specifically, we pick a target m and find two disjoint vertex sets U and W, where U is a linear-sized random set with $e(U) \approx m$ and W is an augmenting set of size $O(\sqrt{n})$ selected from a candidate set $W_0 \subseteq V(G) \setminus U$ whose degrees are of linear order and all lie in a narrow interval. We then show that the subsets $Z \subseteq W$ produce $\Omega(n^{3/2})$ distinct values of $e(U \cup Z)$ that all lie in an interval of length $O(n^{3/2})$ centred at $e(U)$. Applying this to $\Omega(\sqrt{n})$ well-separated targets m yields the full $\Omega(n^2)$ count.

Kwan and Sudakov construct W as three disjoint parts $\mathbf{S}$, $\mathbf{T}$, $\mathbf{X}$, each of size $\Theta(\sqrt{n})$, that act at complementary scales. Every element has $\Theta(n)$ edges into U, and the degrees from $\mathbf{S}$ into U are uniformly smaller than those from $\mathbf{T}$ by a gap of $\Omega(\sqrt{n})$. Varying the number k of elements of $\mathbf{S}$ included in Z shifts $e(U \cup Z)$ at the coarsest scale $\Theta(n)$. Exchanging elements of $\mathbf{S}$ for elements of $\mathbf{T}$ shifts it by $\Theta(\sqrt{n})$. The elements of $\mathbf{X}$ fill the remaining $\Theta(\sqrt{n})$ gap by contributing $\Theta(\sqrt{n})$ distinct degrees

into $U \cup Z$ for each choice of Z. Hence the different combinations of $\mathbf{S}, \mathbf{T}, \mathbf{X}$ yield $\Omega(n^{3/2})$ distinct values of $e(U \cup Z)$.

Achieving the last requirement that $\mathbf{X}$ provides $\Theta(\sqrt{n})$ distinct degrees into $U \cup Z$ for each choice of Z is the main technical challenge, as vertex diversity defined in Chapter 2 does not produce enough distinct degrees in this case. In particular, by the anticoncentration principle from Section 1.1, we have $\mathbb{P}[d_U(x) = d_U(y)] = O(1/\sqrt{n})$ for all but at most n^δ pairs $x, y \in V(G)$ in an (ε, δ)-diverse graph. Following the same construction as in the proof of Lemma **??**, the collision graph H on the candidate set W_0 has expected number of edges

$$\mathbb{E}[e(H)] = O(|W_0|^2/\sqrt{n} + |W_0| \cdot n^\delta),$$

which yields an independent set $I \subseteq W_0$ of size

$$|I| = \frac{|W_0|^2}{2\mathbb{E}[e(H)] + |W_0|} = \Omega\left(\sqrt{n} \cdot \frac{|W_0|}{|W_0| + n^{1/2+\delta}}\right).$$

by Turán's theorem (Proposition 1.5). Thus for $|I|$ to reach $\Omega(\sqrt{n})$, we need $|W_0| = \Omega(n^{1/2+\delta})$. Unfortunately, W_0 is extracted by pigeonholing n vertices into $O(\sqrt{n})$ degree intervals of length $O(\sqrt{n})$, so at best we have $|W_0| = \Theta(\sqrt{n})$, yielding only $|I| = \Omega(n^{1/2-\delta})$.

To overcome this obstacle, Kwan and Sudakov's relaxes the canditate set W_0 to also contain disjoint *pairs*. Write $\mathbf{x}$ in bold denote an object that is either a single vertex or a pair of vertices, and similarly for sets of such objects. For $\mathbf{x} = \{x_1, x_2\}$, set $N(\mathbf{x}) := N(x_1) \cup N(x_2)$ as a multiset union and $d(\mathbf{x}) := |N(\mathbf{x})| = d(x_1) + d(x_2)$ as the pair-degree, and note that there are $\binom{n}{2} = \Theta(n^2)$ pairs but the pair-degrees take only $O(n)$ values. Pigeonhole now yields $\Theta(n)$ disjoint pairs sharing a common pair-degree, vastly exceeding the $n^{1/2+\delta}$ threshold. The price is that we need the symmetric difference $|N(\mathbf{x})\Delta N(\mathbf{y})|$ to be $\Omega(n)$ not just for most pairs of vertices, but for most pairs of *pairs*. We call this property *pair-diversity*.

Definition 3.1. An n-vertex graph G is $(\varepsilon, \delta, \alpha)_2$-*diverse* if for each pair of vertices $\mathbf{x} = \{x_1, x_2\}$ in G with $|N(x_1)\Delta N(x_2)| \geq \alpha n$, there are at most n^δ disjoint pairs of vertices $\mathbf{y} = \{y_1, y_2\}$ with $\mathbf{x} \cap \mathbf{y} = \emptyset$, such that $|N(\mathbf{x})\Delta N(\mathbf{y})| < \varepsilon n$.

It turns out that richness not only implies diversity, but also pair-diversity.

Lemma 3.3. *Let an n-vertex graph G be (ε, δ)-rich, with $\delta \leq 1/2$. Then G is $(\varepsilon, \delta, \alpha)_2$-diverse for any $\alpha \geq 2\delta$.*

Proof. Suppose not. Then there exists a pair $\mathbf{x} = \{x_1, x_2\}$ with $|N(x_1)\Delta\overline{N(x_2)}| \geq \alpha n$ and a set Y of n^δ pairs such that $|N(\mathbf{x})\Delta N(\mathbf{y})| < \alpha\varepsilon n/2$ for all $\mathbf{y} \in Y$. Since

$$|N(x_1)\Delta\overline{N(x_2)}| = |N(x_1) \cap N(x_2)| + |\overline{N(x_1)} \cap \overline{N(x_2)}| \geq \alpha n,$$

we have $|N(x_1) \cap N(x_2)| \geq (\alpha/2)n \geq \delta n$ or $|\overline{N(x_1)} \cap \overline{N(x_2)}| \geq (\alpha/2)n \geq \delta n$. Assume without loss of generality that the former case holds. Then for each vertex y in any $\mathbf{y} \in Y$,

$$\left|\overline{N(y)} \cap (N(x_1) \cap N(x_2))\right| \leq |N(\mathbf{x}) \Delta N(\mathbf{y})| < \alpha\varepsilon n/2 \leq \varepsilon |N(x_1) \cap N(x_2)|.$$

But then the set of all such y now contradicts the (ε, δ)-rich condition. □

We also introduce a reversed notion of diversity.

Lemma 3.4. *Let an n-vertex graph G be (ε, δ)-rich. Then G has at most $n^{1+\delta}$ pairs of vertices $x_1, x_2 \in V(G)$ with $|N(x_1) \Delta \overline{N(x_2)}| < (\varepsilon/2)n$.*

Proof. We show that for each choice of $x_1 \in V(G)$, there are at most n^δ vertices $x_2 \in V(G)$ with $|N(x_1) \Delta \overline{N(x_2)}| < (\varepsilon/2)n$. Since either $|N(x_1)| \geq n/2$ or $|\overline{N(x_1)}| \geq n/2$, we assume without loss of generality that $|N(x_1)| \geq n/2$. By (ε, δ)-richness, there are at most n^δ vertices x_2 with $|N(x_2) \cap N(x_1)| < \varepsilon n/2$. For all other choices of x_2,

$$|N(x_1) \Delta \overline{N(x_2)}| \geq |N(x_1) \cap N(x_2)| \geq (\varepsilon/2)n.$$

The result now follows from the fact that there are only n choices of x_1. □

Remark: In contrast to diversity, this lemma gives an upper bound on the symmetric difference between the neighbourhoods of two vertices. Specifically, since $|N(x_1) \Delta N(x_2)| + |N(x_1) \Delta \overline{N(x_2)}| = n - 2$ for any two vertices x_1, x_2,

$$|N(x_1) \Delta N(x_2)| \leq (1 - \varepsilon/2)n$$

for most pairs of vertices x_1, x_2.

One last thing we need is to ensure $\mathbf{X}$ actually provides $\Theta(\sqrt{n})$ distinct degrees into $U \cup \mathbf{Z} := U \cup \bigcup_{\mathbf{z} \in \mathbf{Z}} \mathbf{z}$ for each choice of augmenting set $\mathbf{Z} \subseteq \mathbf{S} \cup \mathbf{T}$. Notice that the current construction of $\mathbf{X}$ only ensures distinct degrees $d_U(\mathbf{x})$ into U instead of $d_{U \cup Z}(\mathbf{x})$. If U were sampled once and directly constructed $\mathbf{S}, \mathbf{T}, \mathbf{X}$ based on U, then the $d_U(\mathbf{x})$ would be fixed numbers given the outcome of U. However, there is no guarantee that $d_{U \cup Z}(\mathbf{x}) = d_U(\mathbf{x}) + d_Z(\mathbf{x})$ would remain distinct, as the addition of $d_Z(\mathbf{x})$ could cancel out the uniqueness of $d_U(\mathbf{x})$. Kwan and Sudakov overcome this with a *double-exposure* technique. They perform two rounds of sampling. In the first round, they sampled the random set U_0 and construct $\mathbf{S}, \mathbf{T}, \mathbf{X}$ relative to U_0. In the second round, they sampled U from U_0 by independently keeping each vertex with probability $1/2$. After the first sampling, the construction guarantees $|N_{U_0}(\mathbf{x}) \Delta N_{U_0}(\mathbf{y})| = \Omega(n)$ for all $x, y \in \mathbf{X}$, and the augmenting sets $\mathbf{Z} \subseteq \mathbf{S} \cup \mathbf{T}$ are fully determined by U_0. In the second round of sampling, the difference $d_U(\mathbf{x}) - d_U(\mathbf{y})$ becomes a sum of $\Omega(n)$ independent Bernoulli terms, and Littlewood-Offord yields $\mathbb{P}[d_U(\mathbf{x}) - d_U(\mathbf{y}) = t] = O(1/\sqrt{n})$ for every integer t. In

particular, since $t_{xy} = d_Z(\mathbf{y}) - d_Z(\mathbf{x})$ is a fixed integer for each $\mathbf{x}, \mathbf{y}$, we may bound for the collision probability

$$\mathbb{P}[d_{U\cup Z}(\mathbf{x}) = d_{U\cup Z}(\mathbf{y})] = \mathbb{P}[d_U(\mathbf{x}) - d_U(\mathbf{y}) = t_{xy}] = O(1/\sqrt{n}).$$

The same collision graph construction now finds us the desired set of U such that $d_{U\cup Z}(\mathbf{x})$ is distinct for $\mathbf{x} \in \mathbf{X}$.

3.1 The main lemma

As outlined above, we will find many induced subgraphs $G[U]$ whose edge counts are well-separated from each other, and show that each can be augmented in $\Omega(n^{3/2})$ ways. Theorem 3.2 will be an immediate consequence of the following lemma, which we will dedicate the rest of this chapter to prove.

Lemma 3.5. *Let $C > 0$. Then there exists $c = c(C) > 0$, such that for any C-Ramsey graph G on n vertices and any m satisfying $cn^2 \leq m \leq 2cn^2$, there are disjoint subsets $U, W \subseteq V(G)$ with $|e(U) - m| = O(n^{3/2})$ and $|W| = O(\sqrt{n})$ such that*

$$|\{e(U \cup Z) : Z \subseteq W\}| = \Omega(n^{3/2}).$$

We first show how this implies Theorem 3.2.

Proof of Theorem 3.2. Note that for any U, W given by Lemma 3.5,

$$|e(U \cup W) - e(U)| = O(n^{3/2}),$$

as $|W| = O(\sqrt{n})$ and each $w \in W$ has at most n neighbours in $U \cup W$. But then the $\Omega(n^{3/2})$ values of $e(U \cup Z)$ for $Z \subseteq W$ are contained in an interval of length $O(n^{3/2})$ centered at $e(U)$. Partition the interval between cn^2 and $2cn^2$ into $\frac{c}{k}\sqrt{n}$ subintervals of length $kn^{3/2}$, for some large enough constant k to be determined. Now apply Lemma 3.5 to a suitable choice of m in each subinterval so that the obtained values of $e(U \cup Z)$ for $Z \subseteq W$ lie in the same subinterval as m, which is allowed for sufficiently large k. Since we applied Lemma 3.5 to $\Omega(\sqrt{n})$ disjoint subintervals, we obtain $\Omega(n^2)$ distinct sizes of induced subgraphs. □

3.2 The construction of augmenting sets

We now prove the key construction that produces the sets $U_0, \mathbf{S}, \mathbf{T}, \mathbf{X}$ with the properties needed for Lemma 3.5.

Lemma 3.6. *For any $C > 0$ there exists $c = c(C) > 0$ such that the following holds. For any C-Ramsey graph G on n vertices and any m satisfying $cn^2 \leq m \leq 2cn^2$, there is a vertex set $U_0 \subseteq V(G)$ with $|e(U_0) - 4m| = O(n^{3/2})$, a constant $c' = c'(C) > 0$, and sets $\mathbf{S}$, $\mathbf{T}$, $\mathbf{X}$ of size $c'\sqrt{n}$ such that*

1. *either* $U_0, \mathbf{S}, \mathbf{T}, \mathbf{X} \subseteq V(G)$ *are disjoint sets of vertices, or* $\mathbf{S}, \mathbf{T}, \mathbf{X} \subseteq \binom{V(G)}{2}$ *are sets of disjoint pairs of vertices with no vertex appearing in more than one of* $U_0, \mathbf{S}, \mathbf{T}, \mathbf{X}$,
2. *there is* $d = \Theta(n)$ *such that* $d_{U_0}(\mathbf{x}) = d + O(\sqrt{n})$ *for each* $\mathbf{x} \in \mathbf{S} \cup \mathbf{T} \cup \mathbf{X}$,
3. $\min_{\mathbf{x}\in\mathbf{T}} d_{U_0}(\mathbf{x}) - \max_{\mathbf{x}\in\mathbf{S}} d_{U_0}(\mathbf{x}) \geq 8c'\sqrt{n}$,
4. *for each* $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{X}}{2}$, *we have* $|N_{U_0}(\mathbf{x}) \Delta N_{U_0}(\mathbf{y})| = \Omega(n)$.

Property 2 ensures that each element contributes $\Theta(n)$ edges to any subgraph containing U_0, so that adding or removing a single element shifts the edge count by $\Theta(n)$. Property 3 guarantees that swapping an element of $\mathbf{S}$ for an element of $\mathbf{T}$ shifts the edge count by $\Theta(\sqrt{n})$. Property 4 is the anticoncentration condition on $\mathbf{X}$ that sets up the double-exposure step.

The proof will start with reducing to a rich subgraph of linear size using Lemma 2.4 and using the pigeonhole principle to obtain a large candidate set $\mathbf{W}_0$ of vertices or pairs with similar degrees, satisfying Property 1 and Property 2. We will then sample U_0 at random and verify that all desired properties hold simultaneously with positive probability.

Proof of Lemma 3.6. Let $\varepsilon = \varepsilon(C)$ be given by Lemma 2.4 and set $\delta = \varepsilon/4$. By Lemma 2.4, G contains an (ε, δ)-rich induced subgraph $G[V']$ on $\Omega(n)$ vertices. Since $\log |V'| \geq \frac{1}{2} \log n$, the graph $G[V']$ is $2C$-Ramsey, so by adjusting constants it suffices to work inside $G[V']$. In particular, we may assume G itself is (ε, δ)-rich.

By Lemma **??** with $\alpha = \varepsilon/2$, this richness assumption furnishes the following three properties:

(a) G is $(\varepsilon/2, \delta)$-diverse,

(b) G is $(\varepsilon^2/4, \delta, \varepsilon/2)_2$-diverse,

(c) there are at most $n^{1+\delta}$ pairs $\{x_1, x_2\}$ with $|N(x_1) \Delta \overline{N(x_2)}| < (\varepsilon/2)n$.

Each of the $\binom{n}{2} = \Theta(n^2)$ vertex pairs $\{x_1, x_2\}$ has a pair-degree $d(\{x_1, x_2\}) = d(x_1) + d(x_2)$ taking values from 0 to $2n$, and so by the pigeonhole principle there exists a value d^* and a collection H of $\Omega(n^{3/2})$ pairs with $d(\mathbf{x}) = d^* + O(\sqrt{n})$ for all $\mathbf{x} \in H$. By the remark after Lemma **??**, $d^* = \Theta(n)$. Interpret H as a graph on $V(G)$ whose edges are the $\Omega(n^{3/2})$ pairs. By (c), we may delete $O(n^{1+\delta}) = o(n^{3/2})$ edges in H so that all remaining pairs have $|N(x_1) \Delta \overline{N(x_2)}| \geq (\varepsilon/2)n$. Suppose there exists vertex v in H with degree $\Omega(n^{3/4})$. Take $\mathbf{W}_0 = N_H(v)$ and set $d = d^* - d(v)$. Then each $w \in \mathbf{W}_0$ is a single vertex with degree $d(w) = d + O(\sqrt{n})$. Now suppose H has maximum degree $\Delta = o(n^{3/4})$. By iteratively picking an edge and removing at most 2Δ edges adjacent to it, we obtain a matching of size $\Omega(n^{3/2})/o(n^{3/4}) = \Omega(n^{3/4})$. Take $\mathbf{W}_0$ to be the pairs in this matching and set $d = d^*$. In either case, the subsets of $\mathbf{W}_0$ satisfy Property 1 and Property 2.

Next, we extract a subset $\mathbf{A} \subseteq \mathbf{W}_0$ of size $\Omega(\sqrt{n})$ with pairwise large symmetric differences. Define graph F on $\mathbf{L}$ by joining $\mathbf{x}$ and $\mathbf{y}$ whenever $|N(\mathbf{x})\Delta N(\mathbf{y})| < (\varepsilon^2/4)n$. By diversity, the graph F has $|\mathbf{W}_0| = \Omega(n^{3/4})$ vertices and maximum degree at most $n^\delta < n^{1/4}$. Thus by Turán's theorem, F has an independent set $\mathbf{A} \subseteq \mathbf{W}_0$ with size $\Omega(|\mathbf{W}_0|/n^{1/4}) = \Omega(\sqrt{n})$. That is, for $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$, we have $|N(\mathbf{x})\Delta N(\mathbf{y})| = \Omega(n)$.

By Corollary 1.4 and the Ramsey property, $e(G) \geq 800cn^2$ for some constant $c = c(C) > 0$. For $cn^2 \leq m \leq 2cn^2$, set $p = \sqrt{4m/e(G)}$, so that $p = \Omega(1)$ and $p \leq 0.1$. Let U_0 be a random subset of $V(G)$ formed by including each vertex independently with probability p. We make the following claim:

Claim 3.7. *The following five events each hold with probability greater than* $4/5$*, and therefore all hold simultaneously with positive probability by the union bound.*

(i) $|e(U_0) - 4m| = O(n^{3/2})$,

(ii) There is a set $\mathbf{Q} \subseteq \mathbf{A}$ *involving no vertices of* U_0*, with* $|\mathbf{Q}| \geq (2/3)|\mathbf{A}|$,

(iii) There is a set $\mathbf{R} \subseteq \mathbf{A}$ *with* $|\mathbf{R}| \geq (2/3)|\mathbf{A}|$ *and* $d_{U_0}(\mathbf{x}) = pd + O(\sqrt{n})$ *for each* $\mathbf{x} \in \mathbf{R}$,

(iv) $|N_{U_0}(\mathbf{x})\Delta N_{U_0}(\mathbf{y})| = \Omega(n)$ *for each* $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$,

(v) The equality $d_{U_0}(\mathbf{x}) = d_{U_0}(\mathbf{y})$ *holds for* $O(\sqrt{n})$ *pairs* $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$.

Proof of Claim. We prove each part in order.

(i) Note that $\mathbb{E}[e(U_0)] = 4m$ and there are $O(n^3)$ pairs of edges in G whose presense are dependent in $G[U_0]$, as this only occur if two edges share a vertex. For $e \in E(G)$, let A_e be the event that e is present in $G[U_0]$. Since

$$\mathrm{Var}(e(U_0)) \leq \mathbb{E}[e(U_0)] + \sum_{e \sim e'} \mathbb{P}(A_e \cap A_{e'}) = O(n^3),$$

it follows from Chebyshev's inequality that

$$\mathbb{P}[|e(U_0) - 4m| = \omega(n^{3/2})] \leq \frac{O(n^3)}{\omega(n^3)} \to 0,$$

as $n \to \infty$.

(ii) Let $\mathbf{Q}$ be the subset of $\mathbf{A}$ involving no vertices of U_0. Then $|\mathbf{Q}|$ follows a binomial distribution with $|\mathbf{A}|$ trials, it has mean at least $(1-p)^2|\mathbf{A}| \geq 0.8|\mathbf{A}|$ and variance $O(|\mathbf{A}|)$. It follows from Chebyshev's inequality that

$$\mathbb{P}\left[|\mathbf{Q}| < \frac{2}{3}|\mathbf{A}|\right] \leq \mathbb{P}\left[|\mathbf{Q}| - \mathbb{E}[|\mathbf{Q}|] \geq 0.1|\mathbf{A}|\right] \leq \frac{O(|\mathbf{A}|)}{0.01|\mathbf{A}|^2} = O(1/|\mathbf{A}|) \to 0,$$

as $n \to \infty$.

(iii) For $\mathbf{x} \in |\mathbf{A}|$, the degree $d_{U_0}(\mathbf{x})$ has mean $pd + O(\sqrt{n})$ and variance $O(n)$. By Chebyshev's inequality,

$$\mathbb{P}[d_{U_0}(\mathbf{x}) = pd + O(\sqrt{n})] \leq 1 - \mathbb{P}[|d_{U_0}(\mathbf{x}) - \mathbb{E}[d_{U_0}(\mathbf{x})]| = \omega(\sqrt{n})] \leq 1 - \frac{O(n)}{\omega(n)} \to 1.$$

In particular, $d_{U_0}(\mathbf{x}) = pd + O(\sqrt{n})$ with probability 0.99 when n is sufficiently large. Let $\mathbf{R} \subseteq \mathbf{A}$ be the set of $\mathbf{x}$ that satisfies this bound. Since $\mathbb{E}[|\mathbf{A} \setminus \mathbf{R}|] = 0.01|\mathbf{A}|$. It now follows from Markov's inequality that

$$\mathbb{P}\left[|\mathbf{R}| \geq \frac{2}{3}|\mathbf{A}|\right] = 1 - \mathbb{P}\left[|\mathbf{A} \setminus \mathbf{R}| \geq \frac{1}{3}|\mathbf{A}|\right] \geq 1 - \frac{\mathbb{E}[|\mathbf{A} \setminus \mathbf{R}|]}{\frac{1}{3}|\mathbf{A}|} \geq 0.8.$$

(iv) Recall that $|N(\mathbf{x})\Delta N(\mathbf{y})| = \Omega(n)$ for $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$. Since $|N_{U_0}(\mathbf{x})\Delta N_{U_0}(\mathbf{y})| = |N(\mathbf{x})\Delta N(\mathbf{y}) \cap U_0|$, it has a binomial distribution with parameters $N_{U_0}(\mathbf{x})$ and p. By the Chernoff bound,

$$\mathbb{P}\left[|N_{U_0}(\mathbf{x})\Delta N_{U_0}(\mathbf{y})| < \frac{p}{2}|N(\mathbf{x})\Delta N(\mathbf{y})|\right] = e^{-\Omega(n)} = o(n)$$

The result now follows from the union bound.

(v) For $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$, note that the random variable $d_{U_0}(\mathbf{x}) - d_{U_0}(\mathbf{y})$ is of $(|N((x))\Delta N((y)), p|)$-Littlewood-Offord type. Since $|N(\mathbf{x})\Delta N(\mathbf{y})| = \Omega(n)$, we have $\mathbb{P}[d_{U_0}(\mathbf{x}) = d_{U_0}(\mathbf{y})] = O(1/\sqrt{n})$, and so the expected number of paris $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{A}}{2}$ satisfying $d_{U_0}(\mathbf{x}) = d_{U_0}(\mathbf{y})$ is thus $O(\sqrt{n})$. The result now follows from Markov's inequality.

□

Fix an outcome of U_0 for which all five events hold. Let $\mathbf{A}' = \mathbf{Q} \cap \mathbf{R}$, and note that $|\mathbf{A}'| \geq |\mathbf{A}|/3 = \Omega(\sqrt{n})$. It remains to obtain $\mathbf{S}$, $\mathbf{T}$, and $\mathbf{X}$ from $\mathbf{A}'$ that satisfy Properties 2–4. Consider the graph on $\mathbf{A}'$ where $\mathbf{x}$ and $\mathbf{y}$ are adjacent if and only if $d_{U_0}(\mathbf{x}) = d_{U_0}(\mathbf{y})$. By Turán Theorem (Proposition 1.5), there is an independent set $\mathbf{W} \subseteq \mathbf{A}'$ of size $\Omega(\sqrt{n})$, and note that $\mathbf{W}$ has distinct degrees into U_0. Partition $\mathbf{W}$ into three equal parts by the degree $d_{U_0}(\mathbf{x})$ into U_0, and let $\mathbf{T}$ be the top third, $\mathbf{S}$ be the bottom third, and $\mathbf{X}$ be the middle third. Since $\mathbf{W}$ has distinct degrees and $|\mathbf{X}| = \Omega(\sqrt{n})$, we may find constant c' that satisfies Property 3. Since $\mathbf{X} \subseteq \mathbf{A}$, Property 4 holds. Trimming the sizes of $\mathbf{S}, \mathbf{T}, \mathbf{X}$ to $c'\sqrt{n}$ completes the proof. □

Remark: Claim **??** is where the anticoncentration principle from Section 1.2 enters the proof. For each pair $\mathbf{x}, \mathbf{y} \in \mathbf{A}$ with $|N(\mathbf{x})\Delta N(\mathbf{y})| = \Omega(n)$, the difference $d_{U_0}(\mathbf{x}) - d_{U_0}(\mathbf{y})$ is a sum of $\Omega(n)$ independent bounded random variables (since U_0 is formed by independent Bernoulli sampling). By the Erdős–Littlewood–Offord theorem, the probability that this sum equals zero is $O(1/\sqrt{n})$. This is the Bernoulli-sampling case of the anticoncentration mechanism discussed in Section 1.2, in contrast to the hypergeometric case used in Lemma 2.2.

3.3 The double-exposure argument

Given $\mathbf{S}, \mathbf{T}, c'$ from Lemma 3.6, fix an ordering on $\mathbf{S}$ and $\mathbf{T}$ and define

$$\mathcal{P} := \{(k,i) : c'\sqrt{n} \le k \le 2c'\sqrt{n} \text{ and } 0 \le i \le c'\sqrt{n}\}.$$

For $(k,i) \in \mathcal{P}$, define $\mathbf{Z}_{k,i} \subseteq \mathbf{S} \sqcup \mathbf{T}$ by taking the first $k-i$ elements from $\mathbf{S}$ and the first i elements from $\mathbf{T}$. By Property 2 and 3 of Lemma 3.6, the values of $e(U_0, \mathbf{Z}_{k,i}) := \sum_{\mathbf{z} \in \mathbf{Z}_{k,i}} d_{U_0}(\mathbf{z})$ are evenly spaced by $\Theta(\sqrt{n})$ as (k,i) vary lexicographically, which already forms $\Omega(n)$ distinct values of $e(U_0 \cup \mathbf{Z}_{k,i})$. What remains is to fill the $\Theta(\sqrt{n})$ gaps between each $e(U_0 \cup \mathbf{Z}_{k,i})$ with $\mathbf{X}$ via the *double-exposure* technique while maintaining the nice gap struture of the subsets $\mathbf{Z}_{k,i}$. More specifically, we sample U from U_0 such that the values $e(U \cup \mathbf{Z}_{k,i} \cup \{\mathbf{x}\})$ are distinct as $\mathbf{Z}_{k,i} \subseteq \mathbf{S} \sqcup \mathbf{T}$ and $\mathbf{x} \in \mathbf{X}$ vary. Define $U_{k,i} := U \cup \mathbf{Z}_{k,i}$ and $e_{k,i} := e(U_{k,i}) - e(U) = e(\mathbf{Z}_{k,0}) + e(\mathbf{Z}_{k,0}, U)$. There are four things we need to ensure:

1. *Degree distinctness in* $\mathbf{X}$. For most pairs (k,i), the elements of $\mathbf{X}$ should have $\Omega(\sqrt{n})$ pairwise distinct degrees into $U_{k,i}$, all lying in an interval of length $O(\sqrt{n})$. This is the payoff of double-exposure.

2. *Concentration of the edge count.* The value $e_{k,0}$ should concentrate around its expectation within $O(n)$, so that the $\Theta(n)$ spacing between $e_{k,0}$ and $e_{k+1,0}$ are perserved.

3. *Regular medium-scale spacing.* For each k, the values $e_{k,0}, e_{k,1}, \ldots, e_{k,c'\sqrt{n}}$ should span a range of $\Omega(n)$ and increase roughly uniformly, so that $\Theta(\sqrt{n})$ of them are pairwise separated by $\Omega(\sqrt{n})$.

Write $\Delta_{k,i} := e_{k,i} - e_{k,i-1}$ and we have the following lemma, whose first two properties correspond to the first two points above, and the last two properties combined ensures regular medium-scale spacing.

Lemma 3.8. *Let G be a C-Ramsey graph, let $\mathbf{S}$, $\mathbf{T}$, $\mathbf{X}$, U_0, c', d be obtained from Lemma 3.6. Let U be a random subset of U_0 where each vertex is present with probability $1/2$, and let $U_{k,i}, e_{k,i}, \Delta_{k,i}$ be defined as above. Then there exist positive constants $M, \beta, Q > 0$ depending on C, such that $Q \ge 3\beta$ and for each $c'\sqrt{n} \le k \le 2c'\sqrt{n}$ the following hold:*

1. *With probability at least 0.99, there is a set $\mathcal{I}_k$ of $(1-\beta/(2M)) \cdot c'\sqrt{n}$ values with the following property: for each $i \in \mathcal{I}_k$, there exists $\mathbf{X}_{k,i} \subseteq \mathbf{X}$ of size $\Omega(\sqrt{n})$, such that the values $d_{U_{k,i}}(\mathbf{x})$ for $\mathbf{x} \in \mathbf{X}_{k,i}$ are distinct and all lie in the interval $[d/2 - Q\sqrt{n},\ d/2 + Q\sqrt{n}]$.*

2. *With probability at least 0.99,*
$$|e_{k,0} - \mathbb{E}[e_{k,0}]| \le Qn.$$

3. *With probability at least $1/2$,*
$$e_{k,c'\sqrt{n}} - e_{k,0} \ge 3\beta n.$$

4. *With probability at least* 0.99*,*

$$\sum_{i\,:\,|\Delta_{k,i}|\geq M\sqrt{n}} |\Delta_{k,i}| \leq \beta n.$$

That is, the unusually large increments contribute little to the total range $e_{k,c'\sqrt{n}} - e_{k,0}$.

Proof. We prove each property in the order of 3, 4, 1, 2, so that the constants β, M, Q may be determined in their dependency order.

3. Note that $|e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,i-1})| \leq 2c'\sqrt{n}$, as $|\mathbf{Z}_{k,i}| = k \leq 2c'\sqrt{n}$. This and the construction of U yields

$$\begin{aligned}\mathbb{E}[e_{k,i} - e_{k,i-1}] &= \mathbb{E}[e(U, \mathbf{Z}_{k,i}) - e(U, \mathbf{Z}_{k,i-1})] + [e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,i-1})] \\ &\geq \frac{1}{2}[e(U_0, \mathbf{Z}_{k,i}) - e(U_0, \mathbf{Z}_{k,i-1})] - 2c'\sqrt{n}.\end{aligned}$$

It follows form Property 3 of Lemma 3.6 that $e(U_0, \mathbf{Z}_{k,i}) - e(U_0, \mathbf{Z}_{k,i-1}) \geq 8c'\sqrt{n}$, and so

$$\mathbb{E}[e_{k,i} - e_{k,i-1}] \geq 2c'\sqrt{n},$$

for all i. Put $\beta = 2(c')^2/3$ so the difference $e_{k,c'\sqrt{n}} - e_{k,0}$ has expectation at least $3\beta n$. But then since $e(U, \mathbf{Z}_{k,i}) = \sum_{u\in U_0} \mathbb{1}_{u\in U} \cdot d_{\mathbf{Z}_{k,i}}(u)$,

$$\begin{aligned}e_{k,c'\sqrt{n}} - e_{k,0} &= [e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,0})] + [e(U, \mathbf{Z}_{k,i}) - e(U, \mathbf{Z}_{k,0})] \\ &= [e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,0})] + \sum_{u\in U_0} \mathbb{1}_{u\in U} \cdot (d_{\mathbf{Z}_{k,i}}(u) - d_{\mathbf{Z}_{k,0}}(u))\end{aligned}$$

is of Littlewood-Offord type with probability 1/2, which is symmetrically distributed around its expectation. The result now follows.

4. Note that

$$\Delta_{k,i} = [e(\mathbf{Z}_{k,i}) - e(\mathbf{Z}_{k,i-1})] + \sum_{u\in U_0} \mathbb{1}_{u\in U} \cdot (d_{\mathbf{Z}_{k,i}}(u) - d_{\mathbf{Z}_{k,i-1}}(u))$$

is of Littlewood-Offord type and measures the difference of edge count after swapping exactly one vertex from $\mathbf{S}$ to $\mathbf{T}$, so by Lemma 3.6 the expectation of $\Delta_{k,i}$ is $O(\sqrt{n})$ and each $u \in U_0$ contributes at most $|d_{\mathbf{Z}_{k,i}}(u) - d_{\mathbf{Z}_{k,i-1}}(u)| \leq 2$ to $\Delta_{k,i}$. Thus, the Hoeffding's inequality yields

$$\mathbb{P}[|\Delta_{k,i}| \geq t] \leq 2\exp\left(\frac{2(t - \mathbb{E}[|\Delta_{k,i}|])^2}{4|U_0|}\right) = \exp(-\Omega(t^2/n)).$$

Using the identity $\mathbb{E}[X] = \sum_{t\geq 1} \mathbb{P}(X \geq t)$ for any nonnegative integer random variable X, we get

$$\begin{aligned}
\mathbb{E}\left[|\Delta_{k,i}| \cdot \mathbb{1}_{|\Delta_{k,i}|\geq M\sqrt{n}}\right] &= \sum_{t=1}^{\infty} \mathbb{P}\left[|\Delta_{k,i}| \cdot \mathbb{1}_{|\Delta_{k,i}|\geq M\sqrt{n}} \geq t\right] \\
&= M\sqrt{n} \cdot \mathbb{P}\left[|\Delta_{k,i} \geq M\sqrt{n}|\right] + \sum_{t=M\sqrt{n}}^{\infty} \mathbb{P}\left[|\Delta_{k,i}| \geq t\right] \\
&= M\sqrt{n} \cdot \exp(-\Omega(M^2)) + \sum_{t=M\sqrt{n}}^{\infty} \exp(-\Omega(t^2/n)) \\
&= \sqrt{n} \cdot \exp(-\Omega(M^2)),
\end{aligned}$$

uniformly over M. It now follows from Markov's inequality that, for large enough M,

$$\mathbb{P}\left[\sum_{i\,:\,|\Delta_{k,i}|\geq M\sqrt{n}} |\Delta_{k,i}| > \beta n\right] \leq \frac{\sum_{i=1}^{|\mathbf{S}|} \mathbb{E}\left[|\Delta_{k,i}| \cdot \mathbb{1}_{|\Delta_{k,i}|\geq M\sqrt{n}}\right]}{\beta n} \leq \frac{e^{-\Omega(M^2)}}{\beta} < 0.99.$$

1. For each $\mathbf{x}, \mathbf{y} \in \binom{\mathbf{X}}{2}$, the difference

$$d_U(\mathbf{x}) - d_U(\mathbf{y}) = \sum_{u\in N_U(\mathbf{x})} \mathbb{1}_{u\in U} - \sum_{u\in N_U(\mathbf{y})} \mathbb{1}_{u\in U} = \sum_{u\in N_U(\mathbf{x})\Delta N_U(\mathbf{y})} a_u \mathbb{1}_{u\in U}$$

with $a_u \in \{1, -1\}$. Thus for each k, i, the random variable $d_{U_{k,i}}(\mathbf{x}) - d_{U_{k,i}}(\mathbf{y})$ is of $(|N_U(\mathbf{x})\Delta N_U(\mathbf{y})|, 1/2)$-Littlewood-Offord type, as it is simply translation of $d_U(\mathbf{x}) - d_U(\mathbf{y})$ by some deterministic value. It follows from Property 4 of Lemma 3.6 and Littlewood-Offord theorem (Proposition 1.8) that

$$\mathbb{P}[d_{U_{k,i}}(\mathbf{x}) = d_{U_{k,i}}(\mathbf{y})] = O(1/\sqrt{n}).$$

Let $H_{k,i}$ be the graph on $\mathbf{X}$ with edges $\{\mathbf{x}, \mathbf{y}\} \in \binom{\mathbf{X}}{2}$ such that $d_{U_{k,i}}(\mathbf{x}) = d_{U_{k,i}}(\mathbf{y})$, so we have $\mathbb{E}[e(H_{k,i})] = |\mathbf{X}|^2 \cdot O(1/\sqrt{n}) = O(\sqrt{n})$. By Markov's inequality, for large enough n,

$$\mathbb{P}[e(H_{k,i}) = O(\sqrt{n})] \geq 1 - \frac{\mathbb{E}[e(H_{k,i})]}{\omega(\sqrt{n})} \geq 1 - \frac{\beta}{400M},$$

in which case Turán's theorem (Proposition 1.5) yields an independent set $\mathbf{Y}_{k,i}$ of size $n/(2O(\sqrt{n}) + \sqrt{n}) = 2\gamma\sqrt{n}$, for some constant $\gamma = \gamma(\beta, M) > 0$. In particular, $d_{U_{k,i}}(\mathbf{x})$ is distinct for $\mathbf{x} \in \mathbf{Y}_{k,i}$. The expected proportion of values of i for which this fails to occur is $\beta/400M$, and by Markov's inequality again, it fails only for $\beta/2M$ proportion of values of i with probability at least $1 - (\beta/400M)/(\beta/2M) = 0.995$, in which case put $\mathcal{I}_k$ as the set of i that succeeds.

By Property 2 of Lemma 3.6 we have $\mathbb{E}[d_U(\mathbf{x})] = d/2 + O(\sqrt{n})$ for each $\mathbf{x} \in \mathbf{X}$. Since $d_U(\mathbf{x})$ follows a binomial distribution with parameters $O(n)$ and $1/2$, we also have $\mathrm{Var}(d_U(\mathbf{x})) = O(n)$. Note that by triangle inequality,

$$|d_U(\mathbf{x}) - d/2| \leq |d_U(\mathbf{x}) - \mathbb{E}[d_U(\mathbf{x})]| + O(\sqrt{n}),$$

and so for sufficiently large Q, Chebyshev's inequality yields

$$\mathbb{P}\left[|d_U(\mathbf{x}) - d/2| > Q\sqrt{n}\right] \leq \frac{\mathrm{Var}(d_U(\mathbf{x}))}{Q^2 n} = \frac{O(1)}{Q^2} < \frac{\gamma}{200}.$$

Let $\mathbf{Y}$ be the set of all $\mathbf{x} \in \mathbf{X}$ such that $|d_U(\mathbf{x}) - d/2| \leq Q\sqrt{n}$. Then each $\mathbf{x} \in \mathbf{X}$ is not present in $\mathbf{Y}$ with probability less than $\gamma/200$. By Markov's inequality,

$$\mathbb{P}\left[|\mathbf{Y}| \geq (1-\gamma)|\mathbf{X}|\right] = 1 - \mathbb{P}\left[|\mathbf{X} \setminus \mathbf{Y}| \geq \gamma|\mathbf{X}|\right] \geq 1 - \frac{\gamma|\mathbf{X}|}{200\gamma|\mathbf{X}|} = 0.995.$$

In other words, with probability at least 0.995, there exists large enough constant Q and a set $\mathbf{Y} \subseteq \mathbf{X}$ of size at least $(1-\gamma)|\mathbf{X}|$ whose elements all lie in the interval $[d/2 - Q\sqrt{n},\ d/2 + Q\sqrt{n}]$.

By the union bound, both above events occur with probability at least 0.99, in which case we take $\mathbf{X}_{k,i} = \mathbf{Y}_{k,i} \cap \mathbf{Y}$ for $i \in \mathcal{I}_k$. Since $|\mathbf{Y}_{k,i}| \geq 2\gamma\sqrt{n}$ and $|\mathbf{Y}| \geq (1-\gamma)|\mathbf{X}|$, we have $|\mathbf{X}_{k,i}| \geq \gamma\sqrt{n}$. This proves Property 1.

2. Since

$$e_{k,i} = e(\mathbf{Z}_{k,i}) + e(U, \mathbf{Z}_{k,i}) = e(\mathbf{Z}_{k,i}) + \sum_{u \in U_0} \mathbb{1}_{u \in U} \cdot d_{\mathbf{Z}_{k,i}}(u),$$

the random variable $e_{k,i}$ is of $(O(n), 1/2)$-Littlewood-Offord type with coefficients $d_{\mathbf{Z}_{k,i}}(u) = O(\sqrt{n})$. It follows that $\mathrm{Var}(e_{k,0}) = O(n^2)$ and by Chebyshev's inequality,

$$\mathbb{P}\left[|e_{k,0} - \mathbb{E}[e_{k,0}]| \leq Qn\right] \geq 1 - \frac{\mathrm{Var}(e_{k,0})}{Q^2 n^2} = 0.99,$$

for large enough Q. We end this proof by noting that enlarging Q does not fail Property 1.

□

3.4 Proof of the main lemma

We now assemble Lemma 3.6 and Lemma 3.8 to prove the main lemma.

Apply Lemma 3.6 to obtain $U_0, \mathbf{S}, \mathbf{T}, \mathbf{X}, c', d$, and define $\mathcal{P}, U, \mathbf{Z}_{k,i}, U_{k,i}$ as in Section 3.3. Note that U is the only source of randomness. We prove the main lemma with W being the set of all vertices in $\mathbf{S}, \mathbf{T}, \mathbf{X}$.

Recall that k ranges over the interval between $c'\sqrt{n}$ and $2c'\sqrt{n}$. For each k, union bound shows that all four properties of Lemma 3.8 hold with probability at least

0.4. Let $\mathcal{K}$ be the set of all such k. Then $\mathbb{E}\left[c'\sqrt{n} - |\mathcal{K}|\right] \leq 0.6\, c'\sqrt{n}$, and Markov's inequality yields

$$\mathbb{P}\left[|\mathcal{K}| \geq 0.2\, c'\sqrt{n}\right] = 1 - \mathbb{P}\left[c'\sqrt{n} - |\mathcal{K}| \geq 0.8\, c'\sqrt{n}\right] \geq 1 - \frac{\mathbb{E}\left[c'\sqrt{n} - |\mathcal{K}|\right]}{0.8\, c'\sqrt{n}} \geq 0.25.$$

We now show that $e(U)$ is close to m with high probability. Note that $|m - \mathbb{E}[e(U)]| = O(n^{3/2})$, as $\mathbb{E}[e(U)] = e(U_0)/4$ and $|e(U_0) - 4m| = O(n^{3/2})$ from Lemma 3.6. Since the presence of edges in $G[U]$ are dependent only if they share a vertex, there are $O(n^3)$ dependent pairs of edges in $G[U]$. Thus we have $\mathrm{Var}(e(U)) \leq O(n^3) + \mathbb{E}[e(U)] = O(n^3)$. Since $|e(U) - m| \leq |e(U) - \mathbb{E}[e(U)]| + |\mathbb{E}[e(U)] - m|$, Chebyshev's inequality yields

$$\mathbb{P}\left[|e(U) - m| = O(n^{3/2})\right] \geq \mathbb{P}\left[|e(U) - \mathbb{E}[e(U)]| = O(n^{3/2})\right] \geq 1 - \frac{\mathrm{Var}(e(U))}{\omega(n^3)} \geq 0.9,$$

for large enough n. Fix an outcome U satisfying both events above, which hold together with probability at least $0.25 + 0.9 - 1 = 0.15$. Let $Q, \beta, M, \mathcal{I}_k$ be given by Lemma 3.8.

For each k, we have

$$\mathbb{E}[e_{k,0}] - \mathbb{E}[e_{k-1,0}] = \frac{1}{2}[e(\mathbf{Z}_{k,0}, U_0) - e(\mathbf{Z}_{k-1,0}, U_0)] + O(\sqrt{n}) = \Theta(n),$$

as $\mathbb{E}[d_U(\mathbf{x})] = d_{U_0}(\mathbf{x})/2 = \Theta(n)$ for $\mathbf{x} \in \mathbf{S}$. We find a subset from $\mathcal{K}$ whose realized values $e_{k,0}$ are well-separated by a linear gap. Specifically, let $\mathcal{K}' \subseteq \mathcal{K}$ contain every qth element of $\mathcal{K}$, for sufficiently large q such that the values of $\mathbb{E}[e_{k,0}]$ are separated by at least $4Qn$ for $k \in \mathcal{K}'$. Then, $|\mathcal{K}'| = \Theta(\sqrt{n})$ and Property 2 of Lemma 3.8 ensures that the values of $e_{k,0}$ are separated by at least $2Qn$ for $k \in \mathcal{K}'$.

We move on to partition the linear gaps between the values of $e_{k,0} : k \in \mathcal{K}'$ into square-root sized gaps. Let $k \in \mathcal{K}'$. By Property 3 of Lemma 3.8, we have $e_{k,c'\sqrt{n}} - e_{k,0} \geq 3\beta n$. Consider the range of integers between $e_{k,0}$ and $e_{k,0} + 3\beta n$. Within this range, we take $2\beta\sqrt{n}/M$ intervals of length $M\sqrt{n}$, each separated by a distance of $M\sqrt{n}/2 = \Omega(\sqrt{n})$. By Property 4 of Lemma 3.8, at most $\beta\sqrt{n}/M$ of these intervals contain no value of $e_{k,i}$. Consider the $\beta\sqrt{n}/M$ intervals that do contain a value of $e_{k,i}$ and pick a representative $e_{k,i}$ from each such interval. Let $\mathcal{I}'_k$ be the set of indices i corresponding to these representatives, and let $\mathcal{I}''_k = \mathcal{I}_k \cap \mathcal{I}'_k$. Then

$$|\mathcal{I}''_k| = |\mathcal{I}'_k| + |\mathcal{I}'_k| - |\mathcal{I}'_k \cup \mathcal{I}_k| \geq \beta\sqrt{n}/M - \beta c'\sqrt{n}/2M = \Omega(\sqrt{n})$$

and the values of $e_{k,i}$ are separated by $\Omega(\sqrt{n})$ for $i \in \mathcal{I}''_k$. Recall that $Q \geq 3\beta$, and so by construction $|e_{k,i} - e_{k,0}| \leq 3\beta n \leq Qn$ for $i \in \mathcal{I}''_k$. In particular, amongst choices of $k \in \mathcal{K}'$ and $\mathcal{I}''_k$, we already have $\Omega(n)$ distinct values of $e(U_{k,i})$ that are separated by $\Omega(\sqrt{n})$. It remains to fill the square-root sized gaps between these values.

In increasing order of these $e(U_{k,i})$, let $\mathcal{P}' \subseteq \{(k,i) : k \in \mathcal{K}', i \in \mathcal{I}''_k\}$ be the corresponding index set of every qth of these values. We enlarge q so that the values

$e(U_{k,i})$ from $\mathcal{P}'$ are separated by at least $2Qn$. For each $(k,i) \in \mathcal{P}'$ we have $i \in \mathcal{I}_k$, so Property 1 of Lemma 3.8 furnishes a set $\mathbf{X}_{k,i}$ such that the values $d_{U_{k,i}}(\mathbf{x})$ are distinct for $\mathbf{x} \in \mathbf{X}_{k,i}$ while lying in a fixed interval of length $2Qn$ centered at $d/2$, where d is independent of k and i. In particular, for each $(k,i) \in \mathcal{P}'$, the set

$$\{e(U_{k,i} \cup \mathbf{x}_{k,i}) : \mathbf{x}_{k,i} \in \mathbf{X}_{k,i}\}$$

lie in an interval of length $2Qn$ centered at $e(U_{k,i}) + d/2$, and so these sets do not overlap among different $(k,i) \in \mathcal{P}'$.

Therefore,

$$|\{e(U \cup Z) : Z \subseteq W\}| \geq |\{e(U_{k,i} \cup \mathbf{x}_{k,i}) : (k,i) \in \mathcal{P}',\, \mathbf{x}_{k,i} \in \mathbf{X}_{k,i}\}| = \Omega(n^{3/2}).$$

□

4 The Erdős–McKay Conjecture

The results of the previous chapters establish increasingly strong conclusions about the induced subgraph statistics of Ramsey graphs. Chapter 2 showed that every C-Ramsey graph on n vertices contains an induced subgraph on $\Omega(n)$ vertices with $\Omega(\sqrt{n})$ distinct degrees, and Chapter 3 improved the count of distinct induced subgraph sizes to $|\Phi(G)| = \Omega(n^2)$. Both proofs rely on the same anticoncentration principle that, when the symmetric difference $|N(x) \triangle N(y)|$ is large, the degree difference $\deg_U(x) - \deg_U(y)$ is a sum of many weakly dependent bounded random variables, and its point probabilities are $O(1/\sqrt{n})$.

The Erdős–McKay conjecture asks for a qualitatively stronger result than $|\Phi(G)| = \Omega(n^2)$. It asserts that $\Phi(G)$ contains every integer in a quadratic-length interval, not merely that it is large. The conjecture was resolved by Kwan, Sah, Sauermann, and Sawhney [10], and this chapter is dedicated to outlining their proof. The approach fundamentally shifts from bounding degree-collision probabilities for pairs of vertices to directly controlling the point probabilities of $e(U)$ itself. This requires a substantially richer set of tools, but the underlying principle remains: when a random variable is approximately normal with large standard deviation, its point probabilities are small.

4.1 From subgraph sizes to edge-count anticoncentration

Assume $V(G) = [n]$ and define the *graph polynomial*

$$f(\xi_1, \ldots, \xi_n) = \sum_{ij \in E(G)} \xi_i \xi_j.$$

Notice that for any $U \subseteq V(G)$, setting $\xi_v = \mathbb{1}_{v \in U}$ gives $f(\vec{\xi}^{(U)}) = e(U)$. The Erdős–McKay conjecture is thus equivalent to asking whether the range of this quadratic polynomial on $\{0,1\}^n$ covers every integer in $\{0, \ldots, \delta n^2\}$.

When U is a p-random subset, the random variable $e(G[U])$ has mean $p^2 e(G) = \Theta(n^2)$. For the variance, note that

$$\mathrm{Var}(e(U)) = \sum_{(e,e') \in E(G)^2} \mathrm{Cov}(\mathbb{1}_{e \subseteq U}, \mathbb{1}_{e' \subseteq U}).$$

The covariance is zero if e and e' share no vertex and $\Theta(1)$ if e and e' share a vertex. For a fixed vertex v, the number of such pairs involving v is $\binom{\deg(v)}{2}$. Since Ramsey graphs have density bounded away from 0 and 1 by Theorem 1.3, it follows from Jensen's inequality that there are

$$\sum_v \binom{\deg(v)}{2} \geq n \binom{2e(G)/n}{2} = \Theta(n^3)$$

pairs of edges that share a vertex, giving standard deviation $\sigma(e(U)) = \Theta(n^{3/2})$.

This variance computation reveals the scale at which the next theorem operates. In Chapters 2 and 3, the anticoncentration analysis targeted the degree difference $\deg_U(x) - \deg_U(y)$, a *linear* polynomial of the vector $(\mathbb{1}_{v\in U})_{v\in V}$ with standard deviation $\Theta(\sqrt{n})$, and the control of $e(U)$ was achieved indirectly through the construction of $\mathbf{S}, \mathbf{T}, \mathbf{X}$. The approach of [10] instead targets the distribution of $e(U)$ directly as a *quadratic* polynomial with standard deviation $\Theta(n^{3/2})$.

Theorem 4.1 (Kwan–Sah–Sauermann–Sawhney [10])**.** *Fix constant $C > 0$. Let G be a C-Ramsey graph on n vertices and let U be a $(1/2)$-random subset of $V(G)$. Then*

$$\sup_{x\in\mathbb{Z}} \mathbb{P}[e(U) = x] = O(n^{-3/2}),$$

and for every fixed $A > 0$,

$$\inf_{\substack{x\in\mathbb{Z} \\ |x-e(U)/4|\le An^{3/2}}} \mathbb{P}[e(U) = x] = \Omega(n^{-3/2}).$$

In other words, the point probabilities of $e(U)$ are all of order $\Theta(1/\sigma)$ within the bulk of the distribution. This theorem is essentially an approximation of the local limit theorem for $e(U)$ up to constant factors. In particular, a true local CLT would give $\mathbb{P}[e(U) = x] \sim \varphi((x-\mu)/\sigma)/\sigma$, where φ is the standard Gaussian density function.

The Erdős-McKay conjecture follows directly from the lower bound of Theorem 4.1 combined with an earlier result on small values of x by Alon, Krivelevich, and Sudakov.

Theorem 4.2 (Erdős–McKay conjecture [10])**.** *Fix $C > 0$, and let G be a C-Ramsey graph on n vertices for sufficiently large n. Then there exists $\delta = \delta(C) > 0$ such that for any integer $0 \le x \le \delta e(G)$, there exists an induced subgraph with x edges.*

Proof. Alon, Krivelevich, and Sudakov [1] proved that there exists $\alpha = \alpha(C) > 0$ such that the theorem holds for all $0 \le x \le n^\alpha$. Thus we may assume $x \ge n^\alpha$. Let m be the smallest integer such that $e(\{1,\dots,m\}) \ge x/4$ and put $G' = G[\{1,\dots,m\}]$. Then

$$e(G') \ge x/4 \ge e(\{1,\dots,m-1\}) \ge e(G') - m,$$

so $|x - e(G')/4| \le m/4 \le m^{3/2}$. Since $m^2 \ge e(G') \ge x/4 \ge n^\alpha$, we have $m \ge n^{\alpha/2}$, making G' a $(2C/\alpha)$-Ramsey graph. Applying Theorem 4.1 to G' with $A = 1$ gives

$$\mathbb{P}[e(G'[U]) = x] = \Omega(m^{-3/2}) = \Omega(n^{3\alpha/4}) > 0.$$

Therefore, there exists $U \subseteq V(G') \subseteq V(G)$ such that $e(G[U)] = e(G'[U]) = x$, for sufficiently large n. □

4.2 Proof overview

The standard way to approach local limit theorems is via Fourier transforms. For a random variable X, the *characteristic function* of X is defined as

$$\varphi_X(\tau) := \mathbb{E}\left[e^{i\tau X}\right] = \int_{-\infty}^{\infty} e^{i\tau X} \cdot \mathbb{P}[X = x]\, dx, \qquad \tau \in \mathbb{R},$$

and note that $|\varphi_X(\tau)| \leq 1$ for all τ. Thus, applying the Fourier inversion formula yields

$$\mathbb{P}[X = x] = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-i\tau x} \varphi_X(\tau)\, d\tau \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |\varphi_X(\tau)|\, d\tau, \tag{3}$$

for integers-valued X. In particular, any upper bound on the integrated $|\varphi_X(\tau)|$ produces an upper bound on the point probability. For the matching lower bound, we may compare φ_X with the characteristic function $\varphi_Z(\tau) = e^{i\tau\mu - \sigma^2\tau^2/2}$ of a Gaussian random variable Z with mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \mathrm{Var}(X)$. The Gaussian density at x is

$$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\tfrac{(x-\mu)^2}{2\sigma^2}\right),$$

which is of order $\Theta(1/\sigma)$ for any x within $O(\sigma)$ of the mean. Hence, if $|\varphi_X(\tau) - \varphi_Z(\tau)|$ is small, then $\mathbb{P}[X = x]$ inherits the same order of magnitude $\Theta(1/\sigma)$ through (3). If a true local limit theorem held for $X = e(U)$, the proof would amount to verifying suitable estimates on $|\varphi_X(\tau)|$ on $[-\pi, \pi]$, and we would be done.

Unfortunately, the actual local limit theorem does not hold for $e(U)$ in general. The distribution of $e(U)$ can fail to be smooth at the unit scale despite the global shape being approximately Gaussian, which results in spikes in the characteristic function and prevents the integrand in (3) from being uniformly small.

The proof of Theorem 4.1 overcomes this obstacle in two stages. The first stage splits into two cases according to whether the degree sequence of G is additively structured or not. The unstructured case can be handled by a standard Fourier-analytic argument without further intervention. The additively structured case is where $|\varphi_X(\tau)|$ splikes, which we will dedicate most of the effort to handle. The two cases then yield the interval estimate

$$\mathbb{P}\left[|X - x| \leq B\right] = \Theta(n^{-3/2}),$$

for some constant $B = B(C)$. The second stage then upgrades it to a pointwise estimate by showing that the probability mass cannot be too unevenly distributed inside the interval via a switching argument.

4.3 Degree sequence dichotomy

We first decompose $X = e(U)$ into a form that separates the linear and quadratic parts of the polynomial. Set $\xi_v = \mathbb{1}_{v \in U}$ and write $x_v = 2\xi_v - 1 \in \{-1, +1\}$, so that

$\xi_v = (1+x_v)/2$. Substituting into the graph polynomial $e(U) = f(\vec{\xi}) = \sum_{ij\in E(G)} \xi_i\xi_j$, we get

$$X = \sum_{ij\in E(G)} \frac{(1+x_i)(1+x_j)}{4} = \underbrace{\frac{e(G)}{4}}_{\mathbb{E}[X]} + \underbrace{\frac{1}{4}\sum_{v\in V(G)} \deg(v)x_v}_{L} + \underbrace{\frac{1}{4}\sum_{uv\in E(G)} x_u x_v}_{Q}, \quad (4)$$

and we denote the linear term as L and the quadratic term as Q. Since $\mathbb{E}[x_v] = 0$ and $\mathbb{E}[x_u x_v] = 0$ for $u \neq v$, we have that $\mathbb{E}[L] = \mathbb{E}[Q] = 0$, the covariance of L and Q is zero, and the constant term $e(G)/4 = \mathbb{E}[X]$. Note that $\mathrm{Var}(L) = \frac{1}{16}\sum_v \deg(v)^2$ and the Erdős-Szemerédi density bound (Theorem 1.3) gives $e(G) = \Theta(n^2)$, the Jensen's inequality and handshake lemma yield

$$\Omega(n^3) = \frac{(2e(G))^2}{n} = \frac{1}{n}\left(\sum_v \deg(v)\right)^2 \leq \sum_v \deg(v)^2 \leq n\sum_v \deg(v) = O(n^3),$$

which shows that $\mathrm{Var}(L) = \Theta(n^3)$. On the other hand, it is straightforward to see that $\mathbb{E}[x_u x_v x_{u'} x_{v'}] = 1$ when $\{u,v\} = \{u',v'\}$ and 0 otherwise, so the Erdős-Szemerédi density bound gives

$$\mathrm{Var}(Q) = \mathbb{E}[Q^2] = \frac{1}{16}\sum_{uv,u'v'\in E(G)} \mathbb{E}[x_u x_v x_{u'} x_{v'}] = \frac{e(G)}{16} = \Theta(n^2).$$

Therefore, the linear part L dominates the total variance, and so we expect L to drive the global behaviour of X, with Q playing a corrective role. Write $d_v := \deg(v)/4$ so that $L = \sum_v d_v x_v$. Since the variables x_v are independent,

$$\varphi_L(\tau) = \mathbb{E}\left[\exp\left(i\tau\sum_v d_v x_v\right)\right] = \prod_{v\in V(G)} \mathbb{E}\left[\exp(i\tau d_v x_v)\right] = \prod_{v\in V(G)} \cos(\tau d_v). \quad (5)$$

The decomposition (4) thus allows us to bound X by bounding the characteristic function of L which factors well.

For $y \in \mathbb{R}$, denote $\|y\| := \mathrm{dist}(y,\mathbb{Z})$ as the distance from y to the nearest integer and note that $0 \leq \|y\| \leq 1/2$. By Jordan's inequality and $1 - t \leq \exp(-t)$ for $t \geq 0$,

$$|\cos(\pi y)| = \cos(\pi\|y\|) = \sqrt{1-\sin^2(\pi\|y\|)} \leq \sqrt{1-4\|y\|^2} \leq 1-2\|y\|^2 \leq \exp(-2\|y\|^2),$$

for $y \in \mathbb{R}$, and so applying to (5) gives

$$|\varphi_L(\tau)| \leq \exp\left(-2\sum_{v\in V(G)} \left\|\frac{\tau d_v}{\pi}\right\|^2\right). \quad (6)$$

In particular, the right-hand side decays exponentially in n unless many of the quantities $\tau d_v/\pi$ are simultaneously close to integers.

The quantity $\sum_v \|\tau d_v/\pi\|^2$ appearing in the exponent of (6) is either large or small, and this dichotomy is the organising principle of the proof of Theorem 4.1. Throughout what follows, let $\nu = \nu(C) > 0$ be a small constant and take the relevant frequency window to be $0 < |\tau| \le \nu$. Either the quantity is large for every τ in the window, which we call the *unstructured case*, or it fails to be large for some τ in the window, which we call the *structured case*. We describe the two cases in turn.

The unstructured case. In the case that

$$\sum_{v \in V(G)} \left\| \frac{\tau d_v}{\pi} \right\|^2 = \omega(\log n)$$

for all τ with $0 < |\tau| \le \nu$, the bound (6) gives the polynomial decay $|\varphi_L(\tau)| \le n^{-\Omega(1)}$ on the window. Since $\mathrm{Var}(Q) = \Theta(n^2)$ is a factor of n smaller than $\mathrm{Var}(L) = \Theta(n^3)$, the quadratic part perturbs φ_X only negligibly on the same range. Esseen's inequality applied to the difference $\varphi_X - \varphi_Z$, where $\varphi_Z(\tau) = \exp(i\mu\tau - \sigma^2\tau^2/2)$ is the Gaussian characteristic function matching the mean and variance of X, then delivers the interval estimate of Theorem **??**. This case admits a local central limit theorem at unit scale essentially by the standard Fourier-analytic route, and we will not say more about it.

The structured case. Here we consider the opposite case that

$$\sum_{v \in V(G)} \left\| \frac{\tau d_v}{\pi} \right\|^2 = O(\log n),$$

for some τ with $0 < |\tau| \le \nu$. This says that for this one frequency, many of the numbers $\tau d_v/\pi$ are simultaneously close to integers, which is a strong arithmetic constraint on the coefficient vector (d_v). Lemma 4.12 of [10] uses this arithmetic constraint to show that for small enough $\gamma > 0$, the vertex set admits a partition

$$V(G) = R \sqcup (B_1 \sqcup \cdots \sqcup B_m),$$

with $|R| \le n^{1-\gamma}$, $|B_j| = \lceil n^{1-2\gamma} \rceil$ for $j \in [m]$, and real numbers $\kappa_1, \ldots, \kappa_m \ge 0$ such that

$$|d_v - \kappa_j| \le n^{1/2+4\gamma},$$

for $j \in [m]$ and $v \in B_j$. In other words, the vertex set thus splits into $m = \Theta(n^{2\gamma})$ many buckets $B_1, \ldots, B_m$ each with degrees clustered around κ_j, together with an exceptional set R of sublinear size. This allows us to write L as

$$L = \sum_{j=1}^{m} \sum_{v \in B_j} d_v x_v + \sum_{v \in R} d_v x_v = \underbrace{\sum_{j=1}^{m} \kappa_j \sum_{v \in B_j} x_v}_{L_{\mathrm{main}}} + \underbrace{\sum_{j=1}^{m} \sum_{v \in B_j} (d_v - \kappa_j) x_v}_{L_{\mathrm{err}}} + \underbrace{\sum_{v \in R} d_v x_v}_{L_R}.$$

Then conditioning on

$$\vec{\Delta} := (|U \cap B_1|, \ldots, |U \cap B_m|, (x_v)_{v \in R})$$

makes L_{main} and L_R deterministic and reduces the conditional variance of L to $\text{Var}(L_{\text{err}} \mid \vec{\Delta}) \leq n^{2+8\gamma}$. The effect of conditioning on $\vec{\Delta}$ can be thought of as conditioning on the approximate value of L which governs the large-scale behaviour of X, allowing us to "zoom in" to see the smaller-scale behaviour of the distribution. In particular, $U \setminus R$ is conditionally uniform over the product of Boolean slices

$$\binom{B_1}{\Delta_1} \times \cdots \times \binom{B_m}{\Delta_m},$$

and what remains in stage 1 is to evaluate the small-ball probability of $L_{\text{err}} + Q$ on this product of Boolean slices.

It is tempting at this point to hope that, after conditioning on $\vec{\Delta}$, $L_{\text{err}} + Q$ satisfies a local CLT of its own, and that the structured case can be resolved by a second round of the Fourier-analytic strategy applied to Q. The following example proves this hope overly optimistic.

Let G be the disjoint union of two independent copies of $G(n/2, 1/2)$, with vertex sets V_1 and V_2. With high probability G is C-Ramsey for some absolute constant C, and every vertex has degree approximately $n/4$. This d_v is approximately $n/16$ for all v, and so all vertices fall into a single bucket $B_1 = V(G)$. Conditioning on $\vec{\Delta}$ amounts to conditioning on the size $|U|$, say $|U| = n/2$.

Write $s = |U \cap V_1|$, so that $|U \cap V_2| = n/2 - s$. Given s, the distribution of $X = e(G[U])$ is centered around is expectation

$$\mathbb{E}[X \mid s] = \frac{1}{2}\binom{s}{2} + \frac{1}{2}\binom{n/2 - s}{2} = \frac{1}{2}(s - n/4)^2 + \frac{n^2}{32} - \frac{n}{8}.$$

The variable s is hypergeometric with mean $n/4$ and variance approximately $n/16$, so by the hypergeometric CLT, $(s - n/4)/(\sqrt{n}/4) \to Z_1 \sim \mathcal{N}(0, 1)$ in distribution. In particular, $(s - n/4)^2$ contributes a term of order n to $X - \mathbb{E}[X]$ with a scaled limit of αZ_1^2 for some constant $\alpha > 0$. The remaining randomness in X comes from the random distribution of the edges in the two copies of $G(n/2, 1/2)$. This is a sum of $\Theta(n^2)$ independent Bernoulli random variables, which also contributes a term of order n has a scaled limit of a Gaussian βZ_2 for some constant $\beta > 0$. Thus, conditioning on $|U| = n/2$, the normalized X converges to

$$\frac{X - \mathbb{E}[X]}{\sigma} \xrightarrow{d} \alpha Z_1^2 + \beta Z_2,$$

which does not converge to a Gaussian distribution. Therefore, in the structured case, the asymptotic conditional distribution of X need not be Gaussian after the bucket conditioning, but we will show it can always be evaluated at independent Gaussians.

4.4 The structured case

The previous section reduced the additively structured case in Stage 1 to a small-ball estimate for $L_{\mathrm{err}} + Q$ on the product of Boolean slices $\binom{B_1}{\Delta_1} \times \cdots \times \binom{B_m}{\Delta_m}$ obtained by conditioning on the bucket statistic $\vec{\Delta}$. The target at this conditional level is

$$\mathbb{P}\left[|X - x| \le B \mid \vec{\Delta}\right] = O(n^{-1}), \tag{7}$$

which, after averaging over $\vec{\Delta}$, yields the interval estimate $\Theta(n^{-3/2})$ of Theorem **??**. The averaging contributes the remaining factor $n^{-1/2}$, reflecting the fact that only a $\Theta(n^{-1/2})$ fraction of $\vec{\Delta}$ values place $\mathbb{E}[X \mid \vec{\Delta}]$ within $O(n)$ of x. As the chi-squared example at the end of Section 4.3 showed, the conditional distribution of X need not be approximately Gaussian, so the Fourier-analytic route of the unstructured case does not close on its own. KSSS bridge this gap by combining three ingredients, which we describe in turn. The proofs occupy Sections 5 through 11 of [10] and we do not reproduce them.

Ingredient 1. The Gaussian invariance principle We first replace the product of Boolean slices by a Gaussian space. The invariance principle of Mossel, O'Donnell and Oleszkiewicz says that a low-degree polynomial of independent low-influence bounded random variables has a distribution close to that of the same polynomial evaluated at independent standard Gaussians. Applied to $L_{\mathrm{err}} + Q$ in the present setting, it produces a quadratic polynomial

$$f(\vec{Z}) \;=\; \vec{Z}^{\top} F\, \vec{Z} \;+\; \vec{f} \cdot \vec{Z} \;+\; f_0, \qquad \vec{Z} = (Z_1, \ldots, Z_n) \sim \mathcal{N}(0, I_n), \tag{8}$$

with a symmetric matrix $F \in \mathbb{R}^{n \times n}$ and a vector $\vec{f} \in \mathbb{R}^n$ derived from the adjacency matrix of G and the bucket partition, such that the characteristic functions of $L_{\mathrm{err}} + Q$ (given $\vec{\Delta}$) and of $f(\vec{Z})$ are close on the frequency range relevant for small-ball transfer. The matching of characteristic functions is carried out in Lemma 11.1 of [10]. The role of this step is to convert a discrete, dependent domain into a continuous, independent one, which allows us to apply the Carbery–Wright theorem described next.

Ingredient 2. The sharpened Carbery-Wright theorem Given the quadratic polynomial $f(\vec{Z})$ from ingredient 1, we now bound $\sup_x \mathbb{P}[|f(\vec{Z}) - x| \le 1]$. In our application $\sigma(f(\vec{Z})) = \Theta(n)$, since the invariance principle preserves the variance and the computation in (4) gave $\mathrm{Var}(Q) = \Theta(n^2)$. The conditional target $O(n^{-1})$ is therefore asking for a small-ball bound of the form $1/\sigma(f(\vec{Z}))$ at unit scale.

The classical result in this direction is the Carbery-Wright theorem, which states that for quadratic polynomial f in independent standard Gaussians and any $\varepsilon > 0$,

$$\sup_{x \in \mathbb{R}} \mathbb{P}\left[|f(\vec{Z}) - x| \le \varepsilon\right] = O\left(\sqrt{\varepsilon / \sigma(f(\vec{Z}))}\right), \tag{9}$$

which is short of our target $O(1/\sigma(f(\vec{Z})))$. The square-root exponent in (9) cannot be improved unconditionally. For example, $f(\vec{Z}) = Z_1^2$ has $\mathbb{P}[|Z_1^2| \le \varepsilon] = \Theta(\sqrt{\varepsilon})$, so

rank-one forms saturate the bound. However, ruling out rank one is not enough on its own either, as $f(\vec{Z}) = Z_1^2 - Z_2^2$ has rank-2 associated matrix and still satisfies $\mathbb{P}[|f(\vec{Z})| \le \varepsilon] = \Theta(\varepsilon \log(1/\varepsilon))$, losing a logarithmic factor. It turns out that forcing the associated matrix to have effective rank at least three yields the desired bound.

Theorem 4.3 (Sharpened Carbery–Wright, [10, Theorem 1.6])**.** *Let $f(\vec{Z}) = \vec{Z}^\top F \vec{Z} + \vec{f} \cdot \vec{Z} + f_0$ be a real quadratic polynomial in independent standard Gaussians, with F a nonzero symmetric matrix. Suppose that for some $\eta > 0$,*

$$\min_{\substack{\widetilde{F} \in \mathbb{R}^{n \times n} \\ \operatorname{rank}(\widetilde{F}) \le 2}} \|F - \widetilde{F}\|_F^2 \ge \eta \|F\|_F^2. \tag{10}$$

Then for $\varepsilon > 0$,

$$\sup_{x \in \mathbb{R}} \mathbb{P}\left[|f(\vec{Z}) - x| \le \varepsilon\right] = O\left(\varepsilon / \sigma(f(\vec{Z}))\right).$$

The hypothesis (10) says that F is a fixed Frobenius-norm proportion away from every matrix of rank at most 2. In particular, it remains to show (10) for the specific matrix F arising from the bucket structure, and this is where the Ramsey hypothesis enters.

Ingredient 3. The robust rank lemma

Lemma 4.4 (Robust rank, [10, Lemma 10.1])**.** *Fix $C > 0$, $0 < \delta < 1$ and $r \in \mathbb{N}$. Let G be a C-Ramsey graph on n vertices with adjacency matrix A, and let $V(G) = I_1 \cup \cdots \cup I_m$ be a partition into equal parts with $n^{\delta/2} \le m \le 2n^\delta$. Then for every matrix $B \in \mathbb{R}^{n \times n}$ with $\operatorname{rank}(B[I_j \times I_k]) \le r$ for all $j, k \in [m]$,*

$$\|A - B\|_F^2 \;=\; \Omega_{C,r,\delta}(n^2).$$

The lemma says that the adjacency matrix of a Ramsey graph is at Frobenius distance $\Omega(n)$ from every matrix whose restriction to each bucket block is low rank. In the application of Theorem 4.3, the parts I_j are the buckets B_j from Section 4.3 (after a minor adjustment to make the sizes equal) and F is a rescaled copy of A restricted away from the exceptional set R. A hypothetical rank-2 approximation $\widetilde{F}$ violating (10) would in particular satisfy $\operatorname{rank}(\widetilde{F}[B_j \times B_k]) \le 2$ on every bucket block, contradicting Lemma 4.4 with $r = 2$ and $\delta = 2\gamma$. Hence (10) holds with some $\eta = \eta(C, \gamma) > 0$.

Lemma 4.4 is the only place in the first stage where the Ramsey hypothesis is used directly, and its proof is the most combinatorial component in this stage. The argument proceeds by discretising the entries of the approximating matrix B to a finite alphabet, observing that a low-rank block forces a small number of distinct row patterns, and applying Ramsey's theorem on an auxiliary graph to extract a monochromatic structure that contradicts the C-Ramsey assumption on G.

To sum it up, applying ingredients 1, 3, and 2 in sequence yields the interval estimate in (7).

4.5 Switching for pointwise control

The first stage delivers the interval estimate

$$\mathbb{P}[|X - x| \le B] = \Theta(n^{-3/2}), \tag{11}$$

but leaves open the worst case where all of this mass sits on a single integer $x' \neq x$ inside the window $[x - B, x + B]$. We now upgrade the interval estimate to a pointwise bound by showing that the mass cannot be too unevenly distributed across the window via a *switching argument* between different possibilities of the set U. In particular, (11) already implies an upper bound on $\mathbb{P}[X = x]$, so it remains to show a lower bound.

For $y \in U$ and $z \notin U$, the *switch* (y, z) replaces U by $U' = (U \setminus \{y\}) \cup \{z\}$. This operation shifts X by

$$s(y, z; U) \coloneqq e(G[U']) - e(G[U]) = \deg_{U\setminus\{y\}}(z) - \deg_U(y).$$

Fix a set $T \subseteq V(G)^2$ of candidate ordered pairs, to be specified later, and let

$$Y_\ell \coloneqq \#\Big\{(y, z) \in T : y \in U,\ z \notin U,\ s(y, z; U) = \ell\Big\}$$

count candidate switches at U that shift X by exactly ℓ. Notice that each candidate switch with $y \in U, z \in U$, and $s(y, z; U) = \ell$ pairs U with the set $U' = (U \setminus \{y\}) \cup \{z\}$, and the switch (z, y) from U' recovers U and shifts X by $-\ell$. In particular, pairing $(U, (y, z))$ with $(U', (z, y))$ in this way is a bijection. Since the uniform measure on $\{0, 1\}^n$ assigns the same probability 2^{-n} to every U, the bijection gives the symmetry identity

$$\mathbb{E}\Big[Y_\ell \cdot \mathbb{1}_{X=x}\Big] = \mathbb{E}\Big[Y_{-\ell} \cdot \mathbb{1}_{X=x+\ell}\Big], \tag{12}$$

for any integer x and ℓ. By Cauchy-Schwarz inequality,

$$\Big(\mathbb{E}\Big[Y_\ell \cdot \mathbb{1}_{X=x}\Big]\Big)^2 \le \mathbb{E}\Big[(Y_\ell^2 \cdot \mathbb{1}_{X=x})^2\Big] \cdot \mathbb{E}\Big[(\mathbb{1}_{X=x})^2\Big].$$

Rearranging and applying (12) yields

$$\mathbb{P}[X = x] \ge \frac{\Big(\mathbb{E}\Big[Y_{-\ell} \cdot \mathbb{1}_{X=x+\ell}\Big]\Big)^2}{\mathbb{E}\Big[Y_\ell^2 \cdot \mathbb{1}_{X=x}\Big]}. \tag{13}$$

The goal now is to find ℓ and the candidate set T so that

$$\mathbb{E}\Big[Y_\ell \cdot \mathbb{1}_{|X-x|\le B}\Big] \gtrsim \frac{|T|}{\sqrt{n}} \cdot n^{-3/2}, \tag{14}$$

$$\mathbb{E}\Big[Y_\ell^2 \cdot \mathbb{1}_{|X-x|\le B}\Big] \lesssim \left(\frac{|T|}{\sqrt{n}}\right)^2 \cdot n^{-3/2}. \tag{15}$$

The two factors have concrete meanings. The factor $n^{-3/2}$ is $\mathbb{P}[|X - x| \le B]$, supplied by the interval estimate (11). The factor $|T|/\sqrt{n}$ represents $\mathbb{E}[Y_\ell]$, which we now

compute. For any pair $(y,z) \in T$, $\mathbb{P}[y \in U, z \notin U] = 1/4$. Conditional on this event, setting $R_w = 2\mathbb{1}_{w\in U} - 1$ and $a_w = 2\mathbb{1}_{w\in N(z)\setminus N(y)} - 1$,

$$\begin{aligned} s(y,z;U) &= \deg_{U\setminus\{y\}}(z) - \deg_U(y) \\ &= \sum_{w\in N(z),\, w\neq y} \mathbb{1}_{w\in U} - \sum_{w\in N(y)} \mathbb{1}_{w\in U} \\ &= \frac{\deg(z)-\deg(y)}{2} + \frac{1}{2}\sum_{w\in N(y)\triangle N(z)} a_w R_w \end{aligned}$$

is a sum of $|N(y)\triangle N(z)|$ independent $\pm$Bernoulli(1/2) contributions plus the deterministic shift $(\deg(z)-\deg(y))/2$. Therefore, if we can find T such that $\deg(z) - \deg(y) = \Theta(\sqrt{n})$ and $|N(y)\triangle N(z)| = \Theta(n)$ for all $(y,z) \in T$, then a local CLT gives

$$\mathbb{P}[s(y,z;U) = \ell \mid y \in U, z \notin U] = \Theta(1/\sqrt{n})$$

for all ℓ in an $O(\sqrt{n})$ window around the mean, which yields

$$\mathbb{E}[Y_\ell] = \frac{1}{4}\sum_{(y,z)\in T} \mathbb{P}[s(y,z;U) = \ell \mid y \in U, z \notin U] = \Theta(|T|/\sqrt{n}).$$

It remains to construct such a set T. Just as in Chapter 3, we can use the richness of Ramsey graphs to ensure $|N(y)\triangle N(z)| = \Theta(n)$. In particular, we apply Lemma **??** with parameter $\varepsilon = \varepsilon(C) > 0$ and a sufficiently small $\delta < \varepsilon$ to obtain a linear size vertex subset S_0 that induces an (ε,δ)-rich subgraph. Applying richness twice then shows that, for all but $O(|S_0|^{1+\delta})$ ordered pairs $(y,z) \in S_0^2$,

$$|(N(y)\setminus N(z)) \cap S_0| \geq \varepsilon^2|S_0|, \qquad |(N(z)\setminus N(y)) \cap S_0| \geq \varepsilon^2|S_0|. \tag{16}$$

More specifically, by richness all but at most $|S_0|^\delta$ vertices $z \in S_0$ satisfy $|N(z)\cap S_0| \geq \varepsilon|S_0|$. Since $|N(z)\cap S_0| \geq \varepsilon|S_0| \geq \delta|S_0|$, applying richness a second time with set $N(z)\cap S_0$ gives

$$|(N(z)\setminus N(y)) \cap S_0| = |\overline{N(y)} \cap (N(z)\cap S_0)| \geq \varepsilon \cdot |N(z)\cap S_0| \geq \varepsilon^2|S_0|$$

for all but a further $|S_0|^\delta$ choices of $y \in S_0$. The bound for $N(y)\setminus N(z)$ follows symmetrically. Combining both bounds in (16) implies $|N(y)\triangle N(z)| = \Theta(n)$ for all but $O(n^{1+\delta})$ ordered pairs $(y,z) \in S_0^2$. Now partitioning the degree sequence into $\sqrt{n}$ intervals and taking S to be the largest bin of vertices from S_0 satisfies the $\deg(z) - \deg(y) = \Theta(\sqrt{n})$ condition. Define T as the set of ordered pairs $(y,z) \in S^2$ with $y \neq z$ that satisfy (16). The first moment estimate (14) now follows.

The second moment bound (15) is more involved. Write $A_{y,z} = \{y \in U,\, z \notin U,\, s(y,z;U) = \ell\}$ and expanding Y_ℓ^2 gives

$$\mathbb{E}\Big[Y_\ell^2 \cdot \mathbb{1}_{|X-x|\leq B}\Big] = \sum_{(y_1,z_1),(y_2,z_2)\in T} \mathbb{P}\Big[A_{y_1,z_1} \cap A_{y_2,z_2} \cap \{|X-x| \leq B\}\Big].$$

We would like bound this from above by $\mathbb{P}[A_{y_1,z_1}] \cdot \mathbb{P}[A_{y_2,z_2}] \cdot \mathbb{P}[|X - x| \leq B]$. This is achieved via (16) and the *dependent random choice* technique – see [10, Lemmas 13.1, 13.4, 13.9] for the details.

Remark. The inequality (13) is stated with the indicator $\mathbb{1}_{X=x}$, while (14) and (15) are stated for the wider event $\{|X - x| \leq B\}$. To bridge the two, Cauchy–Schwarz is applied in [10] a second time to the higher-order moment $\mathbb{E}[Y_{-B}Y_{-B+1} \cdots Y_B \cdot \mathbb{1}_{|X-x|\leq B}]$, which averages over the choice of ℓ. We do not pursue this iteration here, as the conceptual core remains the comparison (13).

Closing the circle

The same large-symmetric-difference condition that powered the collision graph argument in Chapter 3 drives the second-moment estimate here. There, richness lower-bounded $|N(x) \triangle N(y)|$ for every pair in the rich subgraph and forced the small-ball probability of the linear quantity $\deg_U(x) - \deg_U(y)$ to be $O(1/\sqrt{n})$ by treating it as a sum of independent Bernoulli contributions indexed by $N(x) \triangle N(y)$. Here, the same large symmetric difference forces near-independence of two switching events by giving the four neighbourhoods $N(y_i), N(z_i)$ a large pool of U-coordinates over which to vary independently. In both cases a large symmetric difference supplies a large pool of independent coin flips to a quantity we wish to anticoncentrate.

This is the sense in which Theorem 4.1 closes the circle of the thesis. A single anticoncentration principle threads three increasingly delicate arguments. Chapter 2 controls a single degree difference and produces $\Omega(\sqrt{n})$ distinct degrees. Chapter 3 controls degree collisions across many pairs and produces $\Omega(n^2)$ distinct subgraph sizes. The present chapter splits the principle into two complementary roles. The forward Littlewood–Offord direction reappears in the switching argument as the engine of (P1), and the inverse Littlewood–Offord direction enters as the structured-versus-unstructured dichotomy of Section 4.3. The Ramsey hypothesis enters at every layer, through the Erdős–Szemerédi density bound, through richness, through the robust rank of the adjacency matrix, and through the availability of large structured switching sets via dependent random choice.

5 Concluding Remarks

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A Supplementary Proofs